https://www.kaggle.com/datasets/heeraldedhia/bike-buyers

In [238…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, roc_curve, auc
from sklearn.feature_selection import SelectKBest, chi2, f_regression
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import warnings
warnings.filterwarnings('ignore')
```

In [239…

```
from google.colab import drive
import pandas as pd

# Mount Google Drive
drive.mount('/content/drive')

# Define the path to your CSV file
file_path = '/content/drive/My Drive/Customer_Demographics_Bike_Purchase.csv'  # Change this to your file path

# Read the CSV file
df = pd.read_csv(file_path)

# Display the first 5 rows
df.head()
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", forc
e_remount=True).
```

Out[239]:

	ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distance	Region	Age	Purchased Bike
0	12496	Married	Female	40000.0	1.0	Bachelors	Skilled Manual	Yes	0.0	0-1 Miles	Europe	42.0	No
1	24107	Married	Male	30000.0	3.0	Partial College	Clerical	Yes	1.0	0-1 Miles	Europe	43.0	No
2	14177	Married	Male	80000.0	5.0	Partial College	Professional	No	2.0	2-5 Miles	Europe	60.0	No
3	24381	Single	NaN	70000.0	0.0	Bachelors	Professional	Yes	1.0	5-10 Miles	Pacific	41.0	Yes
4	25597	Single	Male	30000.0	0.0	Bachelors	Clerical	No	0.0	0-1 Miles	Europe	36.0	Yes

Data Preprocessing

In [240…

```
print(df['Gender'].value_counts())
```

```
Gender
Male      500
Female    489
Name: count, dtype: int64
```

In [241…

```
print(df['Education'].value_counts())
```

```
Education
Bachelors              306
Partial College        265
High School            179
Graduate Degree        174
Partial High School     76
Name: count, dtype: int64
```

In [242…

```
print(df['Occupation'].value_counts())
```

```
Occupation
Professional      276
Skilled Manual    255
Clerical          177
Management        173
Manual            119
Name: count, dtype: int64
```

In [243…

```
print(df["Marital Status"].value_counts())
```

```
In [243… print(df["Marital Status"].value_counts())
```

```
Marital Status
Married    535
Single     458
Name: count, dtype: int64
```

```
In [244… print(df["Region"].value_counts())
```

```
Region
North America    508
Europe           300
Pacific          192
Name: count, dtype: int64
```

```
In [245… print(df["Purchased Bike"].value_counts())
```

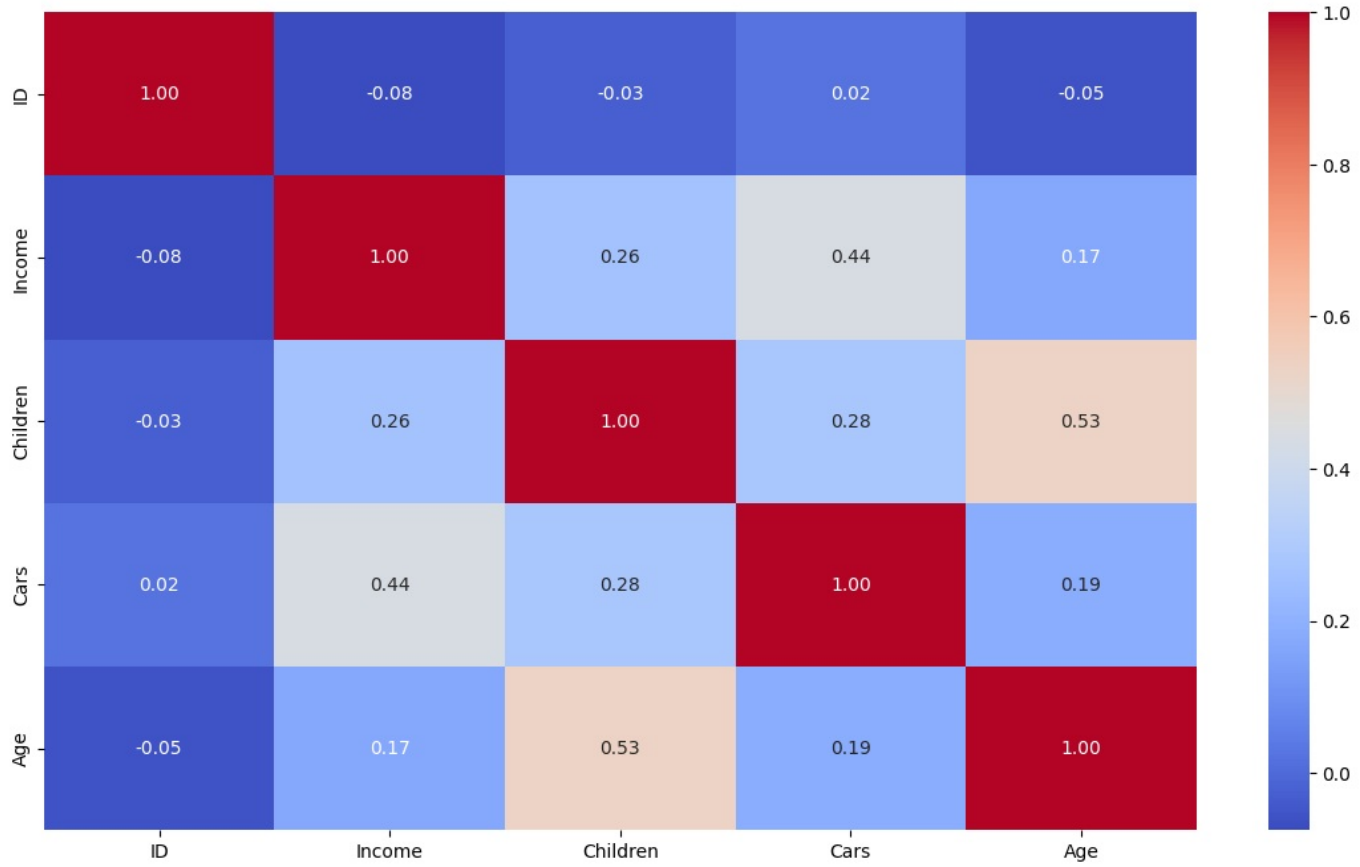
```
Purchased Bike
No     519
Yes    481
Name: count, dtype: int64
```

```
In [246… df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                1000 non-null   int64
 1   Marital Status    993 non-null    object
 2   Gender            989 non-null    object
 3   Income            994 non-null    float64
 4   Children          992 non-null    float64
 5   Education         1000 non-null   object
 6   Occupation        1000 non-null   object
 7   Home Owner        996 non-null    object
 8   Cars              991 non-null    float64
 9   Commute Distance  1000 non-null   object
 10  Region            1000 non-null   object
 11  Age               992 non-null    float64
 12  Purchased Bike    1000 non-null   object
dtypes: float64(4), int64(1), object(8)
memory usage: 101.7+ KB
```

```
In [247… numeric_attribute = df.select_dtypes(include = ['int64', 'float64'])
plt.figure(figsize=(14, 8))
sns.heatmap(numeric_attribute.corr(), annot=True, cmap='coolwarm', fmt='.2f')
```

Out[247]: <Axes: >




```
In [248… ## Finding out correlation of each attribute with each other
```

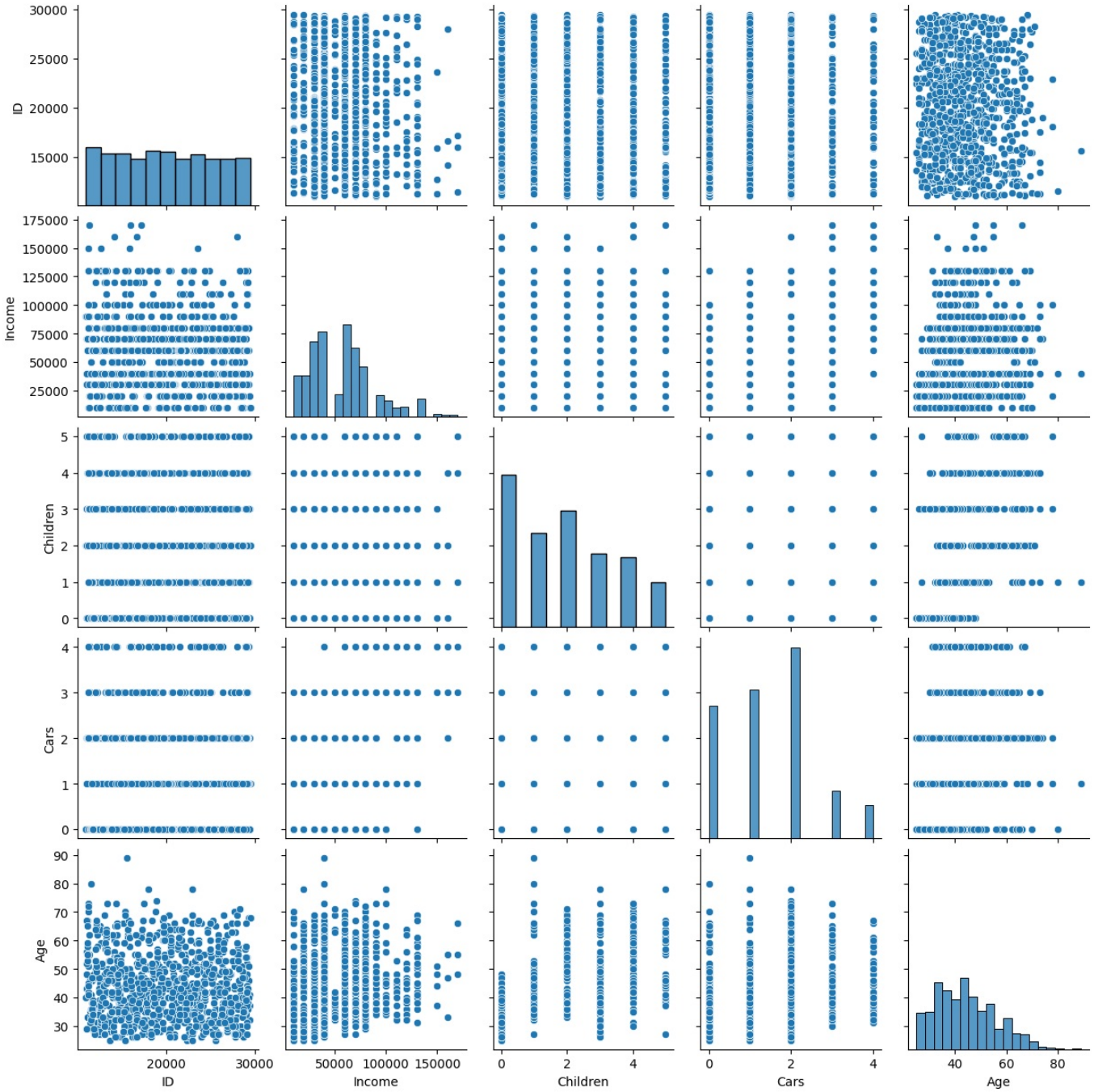
```python
cor_data = numeric_attribute.corr().round(4)
for k in numeric_attribute:
    print(f"\n-------{k}:")
    print(cor_data[k].sort_values(ascending=False))
```

```
-------ID:
ID          1.0000
Cars        0.0221
Children   -0.0287
Age        -0.0542
Income     -0.0751
Name: ID, dtype: float64

-------Income:
Income      1.0000
Cars        0.4400
Children    0.2611
Age         0.1708
ID         -0.0751
Name: Income, dtype: float64

-------Children:
Children    1.0000
Age         0.5317
Cars        0.2802
Income      0.2611
ID         -0.0287
Name: Children, dtype: float64

-------Cars:
Cars        1.0000
Income      0.4400
Children    0.2802
Age         0.1864
ID          0.0221
Name: Cars, dtype: float64

-------Age:
Age         1.0000
Children    0.5317
Cars        0.1864
Income      0.1708
ID         -0.0542
Name: Age, dtype: float64
```

There is no multicollinearity

In [249…
```python
sns.pairplot(df)
```

Out[249]: <seaborn.axisgrid.PairGrid at 0x7fdc3fc2e110>



In [250…

```
## Checking For Duplicate Values
df.duplicated().sum()
```

Out[250]: 0

In [251…

```
## Checking for missing values
df.isnull().sum()
```

Out[251]:

	0
ID	0
Marital Status	7
Gender	11
Income	6
Children	8
Education	0
Occupation	0
Home Owner	4
Cars	9
Commute Distance	0
Region	0
Age	8
Purchased Bike	0

dtype: int64

In [252…

```
df.dropna(inplace=True)
```

In [253…

```
# data after removing missing values
df.isnull().sum()
```

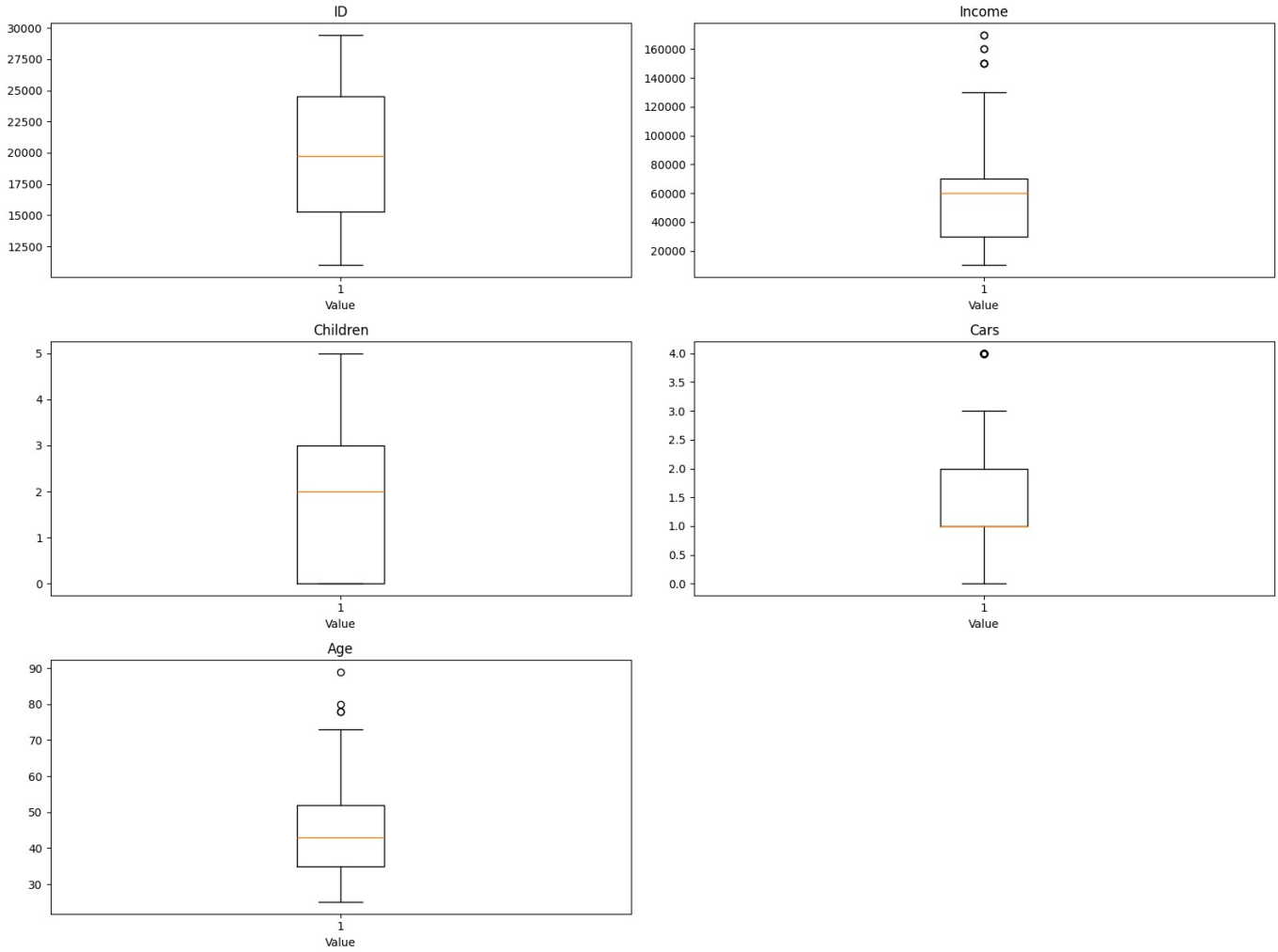
Out[253]:

	0
ID	0
Marital Status	0
Gender	0
Income	0
Children	0
Education	0
Occupation	0
Home Owner	0
Cars	0
Commute Distance	0
Region	0
Age	0
Purchased Bike	0

dtype: int64

In [254…

```
import math
nc = df.select_dtypes(include = ['int64', 'float64'])
# Calculate the number of rows and columns for subplots
num_cols = len(nc.columns)
num_rows = math.ceil(num_cols / 2)  # Adjust the number of columns per row as needed

plt.figure(figsize=(16, 4 * num_rows))

for i, column in enumerate(nc.columns):
    plt.subplot(num_rows, 2, i+1)

    plt.boxplot(nc[column])

    plt.title(column)

    plt.xlabel('Value')
    plt.ylabel('')

plt.tight_layout()
plt.show()
```



In [255…

```
dfb = df.copy()
dfb.head()
```

Out[255]:

	ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distance	Region	Age	Purchased Bike
0	12496	Married	Female	40000.0	1.0	Bachelors	Skilled Manual	Yes	0.0	0-1 Miles	Europe	42.0	No
1	24107	Married	Male	30000.0	3.0	Partial College	Clerical	Yes	1.0	0-1 Miles	Europe	43.0	No
2	14177	Married	Male	80000.0	5.0	Partial College	Professional	No	2.0	2-5 Miles	Europe	60.0	No
4	25597	Single	Male	30000.0	0.0	Bachelors	Clerical	No	0.0	0-1 Miles	Europe	36.0	Yes
5	13507	Married	Female	10000.0	2.0	Partial College	Manual	Yes	0.0	1-2 Miles	Europe	50.0	No

In [256…

```
z_scores = stats.zscore(dfb[['Income', 'Age']])
threshold = 3
outliers = (abs(z_scores) > threshold).any(axis=1)
rows_with_outliers = dfb[outliers]
print("Rows with outliers:")
rows_with_outliers
```

```
Rows with outliers:
```

Out[256]:

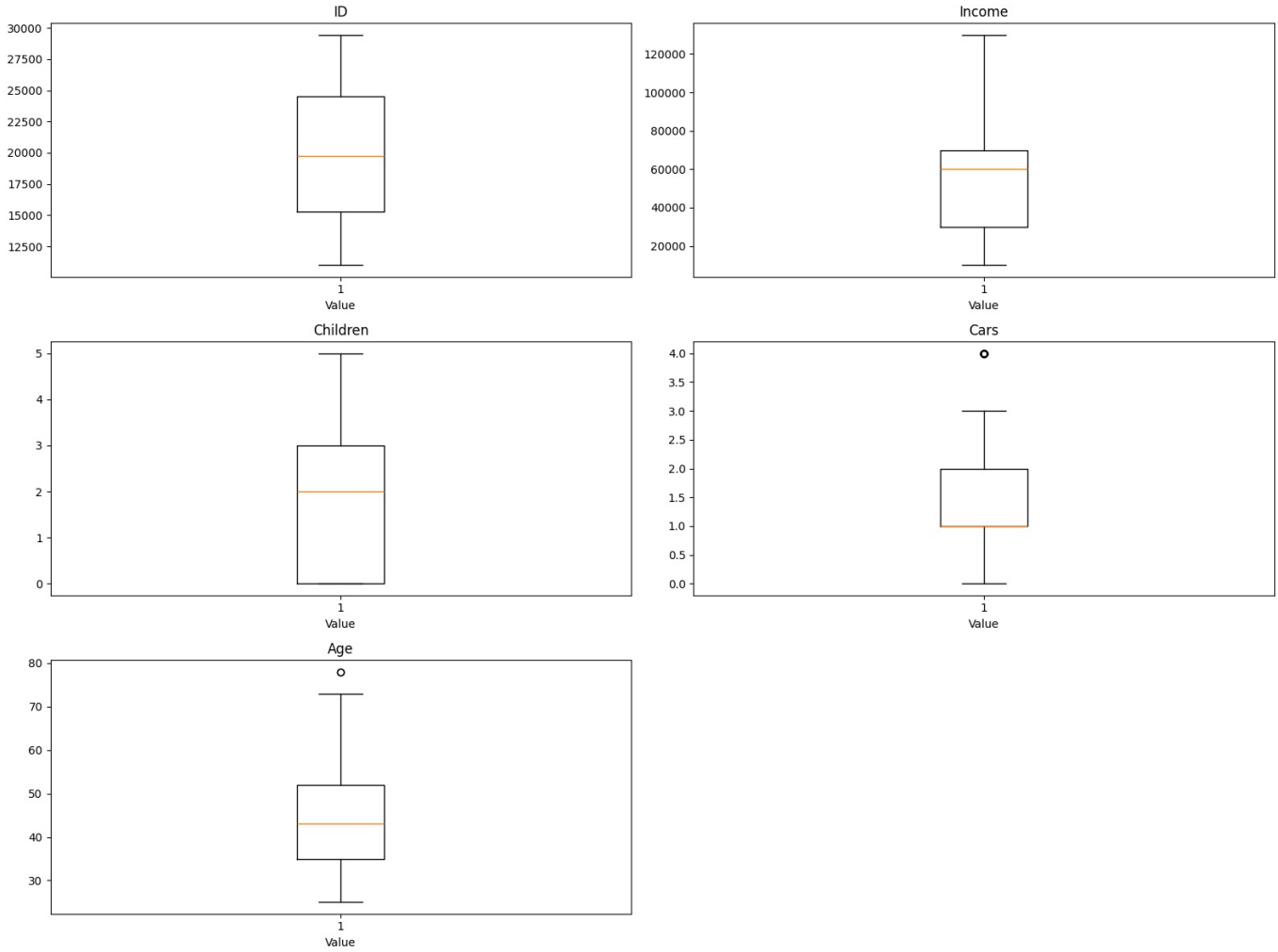
	ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distance	Region	Age	Purchased Bike
43	17185	Married	Female	170000.0	4.0	Partial College	Professional	No	3.0	5-10 Miles	Europe	48.0	Yes
121	15922	Married	Male	150000.0	2.0	High School	Professional	Yes	4.0	0-1 Miles	Europe	48.0	No
178	14191	Married	Male	160000.0	4.0	Partial College	Professional	No	2.0	10+ Miles	Europe	55.0	Yes
259	12705	Married	Male	150000.0	0.0	Bachelors	Management	Yes	4.0	0-1 Miles	Pacific	37.0	Yes
321	16675	Single	Female	160000.0	0.0	Graduate Degree	Management	No	3.0	0-1 Miles	Pacific	47.0	Yes
356	23608	Married	Female	150000.0	3.0	High School	Professional	Yes	3.0	0-1 Miles	Europe	51.0	Yes
375	15628	Married	Female	40000.0	1.0	Bachelors	Skilled Manual	Yes	1.0	0-1 Miles	Europe	89.0	No
401	11555	Married	Female	40000.0	1.0	Bachelors	Clerical	Yes	0.0	0-1 Miles	Europe	80.0	No
829	16009	Single	Male	170000.0	1.0	Graduate Degree	Management	No	4.0	0-1 Miles	North America	66.0	No
993	11292	Single	Male	150000.0	1.0	Partial College	Professional	No	3.0	0-1 Miles	North America	44.0	Yes

In [257…

```
z_scores = stats.zscore(dfb[['Income', 'Age']])
threshold = 3
outliers = (abs(z_scores) > threshold).any(axis=1)
average_income = dfb['Income'].mean()
average_age = dfb['Age'].mean()
dfb.loc[outliers, 'Income'] = average_income
dfb.loc[outliers, 'Age'] = average_age
```

In [258…

```
nc = dfb.select_dtypes(include = ['int64', 'float64'])
# Calculate the number of rows and columns for subplots
num_cols = len(nc.columns)
num_rows = math.ceil(num_cols / 2)  # Adjust the number of columns per row as needed

# Set the figure size
plt.figure(figsize=(16, 4 * num_rows))

# Iterate through each numerical column
for i, column in enumerate(nc.columns):
    plt.subplot(num_rows, 2, i+1)
    plt.boxplot(nc[column])
    plt.title(column)
    plt.xlabel('Value')
    plt.ylabel('')

plt.tight_layout()
plt.show()
```



In [259…
```
dfc = dfb.copy()
dfc.head()
```

Out[259]:

	ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distance	Region	Age	Purchased Bike
0	12496	Married	Female	40000.0	1.0	Bachelors	Skilled Manual	Yes	0.0	0-1 Miles	Europe	42.0	No
1	24107	Married	Male	30000.0	3.0	Partial College	Clerical	Yes	1.0	0-1 Miles	Europe	43.0	No
2	14177	Married	Male	80000.0	5.0	Partial College	Professional	No	2.0	2-5 Miles	Europe	60.0	No
4	25597	Single	Male	30000.0	0.0	Bachelors	Clerical	No	0.0	0-1 Miles	Europe	36.0	Yes
5	13507	Married	Female	10000.0	2.0	Partial College	Manual	Yes	0.0	1-2 Miles	Europe	50.0	No

Label Encoding

In [260…
```
dfc['Gender'] = dfc['Gender'].map({'Male':1, 'Female':0})
dfc['Home Owner'] = dfc['Home Owner'].map({'Yes':1, 'No':0})
dfc['Purchased Bike'] = dfc['Purchased Bike'].map({'Yes':1, 'No':0})
dfc['Marital Status'] = dfc['Marital Status'].map({'Married':1, 'Single':0})
```

In [261…
```
dfc.head()
```

Out[261]:

	ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distance	Region	Age	Purchased Bike
0	12496	1	0	40000.0	1.0	Bachelors	Skilled Manual	1	0.0	0-1 Miles	Europe	42.0	0
1	24107	1	1	30000.0	3.0	Partial College	Clerical	1	1.0	0-1 Miles	Europe	43.0	0
2	14177	1	1	80000.0	5.0	Partial College	Professional	0	2.0	2-5 Miles	Europe	60.0	0
4	25597	0	1	30000.0	0.0	Bachelors	Clerical	0	0.0	0-1 Miles	Europe	36.0	1
5	13507	1	0	10000.0	2.0	Partial College	Manual	1	0.0	1-2 Miles	Europe	50.0	0

In [262…

```
# Define the custom rank mapping
education_mapping = {
    'Partial High School': 0,
    'High School': 1,
    'Partial College': 2,
    'Bachelors': 3,
    'Graduate Degree': 4
}


dfc['Education'] = dfc['Education'].map(education_mapping)
```

In [263…

```
dfc.head()
```

Out[263]:

	ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distance	Region	Age	Purchased Bike
0	12496	1	0	40000.0	1.0	3	Skilled Manual	1	0.0	0-1 Miles	Europe	42.0	0
1	24107	1	1	30000.0	3.0	2	Clerical	1	1.0	0-1 Miles	Europe	43.0	0
2	14177	1	1	80000.0	5.0	2	Professional	0	2.0	2-5 Miles	Europe	60.0	0
4	25597	0	1	30000.0	0.0	3	Clerical	0	0.0	0-1 Miles	Europe	36.0	1
5	13507	1	0	10000.0	2.0	2	Manual	1	0.0	1-2 Miles	Europe	50.0	0

One-hot Encoding

In [264…

```
dfd = dfc.copy()
```

In [265…

```
categorical_cols = ['Occupation', 'Region', 'Commute Distance']
dfd = pd.get_dummies(dfd, columns=categorical_cols, drop_first=True)
dfd = dfd.replace({True: 1, False: 0})
```

In [266…

```
dfd.head()
```

Out[266]:

	ID	Marital Status	Gender	Income	Children	Education	Home Owner	Cars	Age	Purchased Bike	Occupation_Management	Occupation_Ma
0	12496	1	0	40000.0	1.0	3	1	0.0	42.0	0	0	
1	24107	1	1	30000.0	3.0	2	1	1.0	43.0	0	0	
2	14177	1	1	80000.0	5.0	2	0	2.0	60.0	0	0	
4	25597	0	1	30000.0	0.0	3	0	0.0	36.0	1	0	
5	13507	1	0	10000.0	2.0	2	1	0.0	50.0	0	0	

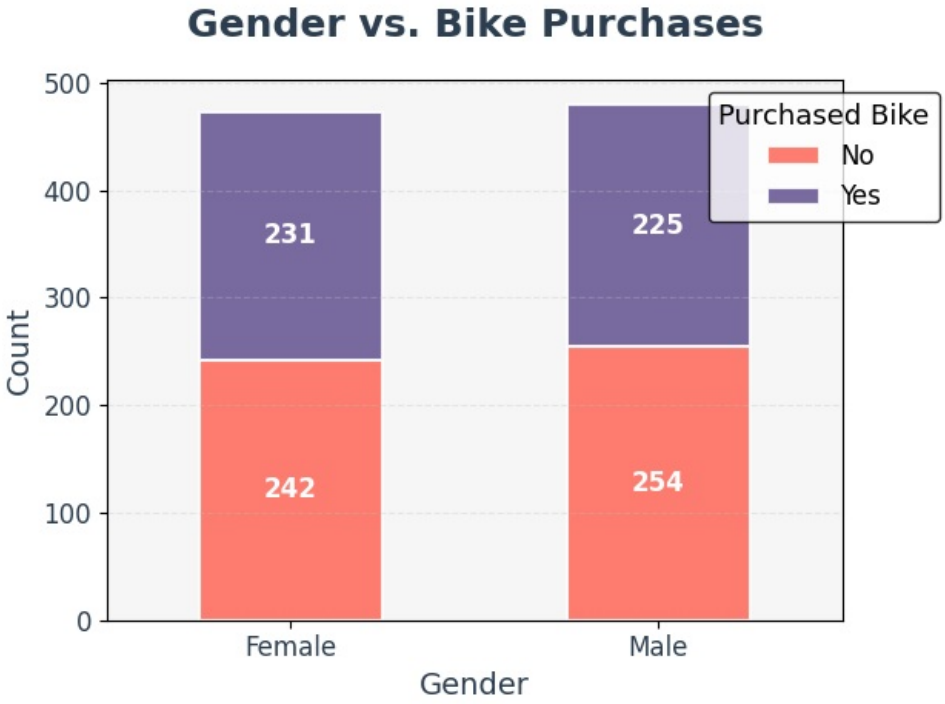
Exploratory Data Analysis

How does Gender relate to bike purchases?

In [268…

```
# Question 2: Gender vs. Bike Purchases
gender_bike = pd.crosstab(df['Gender'], df['Purchased Bike'])

# Create a beautiful stacked bar chart
# plt.figure(figsize=(12, 7))  # Larger figure for elegance
ax = gender_bike.plot(kind='bar', stacked=True,
                      color=['#FF6F61', '#6B5B95'],  # Elegant coral and purple
```

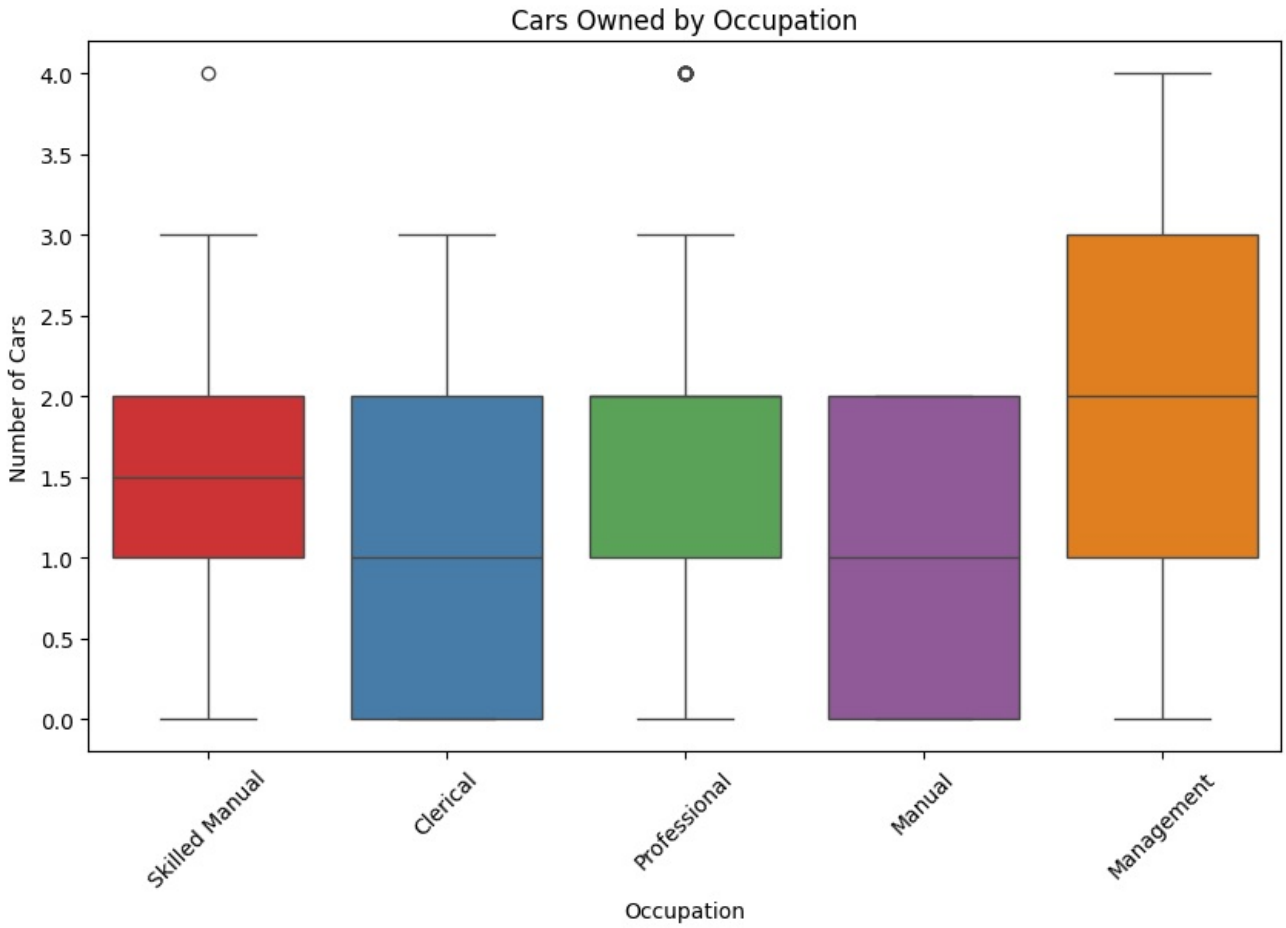
```
                        edgecolor='white', linewidth=1.5,  # White edges for separation
                        alpha=0.9)  # Transparency for depth

# Customize title and labels with elegant fonts
plt.title('Gender vs. Bike Purchases', fontsize=18, fontweight='bold', pad=20, color='#2C3E50')
plt.xlabel('Gender', fontsize=14, color='#2C3E50')
plt.ylabel('Count', fontsize=14, color='#2C3E50')

# Add data labels with a stylish touch
for i in range(len(gender_bike)):
    for j in range(len(gender_bike.columns)):
        count = gender_bike.iloc[i, j]
        total = gender_bike.iloc[i].sum()
        y_position = gender_bike.iloc[i].cumsum()[j] - count/2  # Center label in stack
        if count > 0:  # Only add label if count is non-zero
            plt.text(i, y_position, f'{count}', ha='center', va='center',
                     fontsize=12, fontweight='bold', color='white')

# Customize the legend
plt.legend(title='Purchased Bike', fontsize=12, title_fontsize=13, loc='upper right',
           bbox_to_anchor=(1.15, 1), frameon=True, edgecolor='black')

# Add a subtle grid and background
plt.grid(axis='y', linestyle='--', alpha=0.3, color='#BDC3C7')
plt.gca().set_facecolor('#F5F6F5')  # Light background for contrast
plt.gcf().set_facecolor('#FFFFFF')  # White figure background

# Adjust ticks and layout
plt.xticks(rotation=0, fontsize=12, color='#2C3E50')
plt.yticks(fontsize=12, color='#2C3E50')
plt.tight_layout()

# Save and display
# plt.savefig('gender_bike_purchases.png', dpi=300, bbox_inches='tight')
plt.show()
```



How does occupation influence the number of cars owned?

In [270…

```
# Question 6: Cars Owned by Occupation
plt.figure(figsize=(10, 6))
sns.boxplot(x='Occupation', y='Cars', data=df, palette='Set1')
plt.title('Cars Owned by Occupation')
plt.xlabel('Occupation')
plt.ylabel('Number of Cars')
plt.xticks(rotation=45)
plt.show()
```

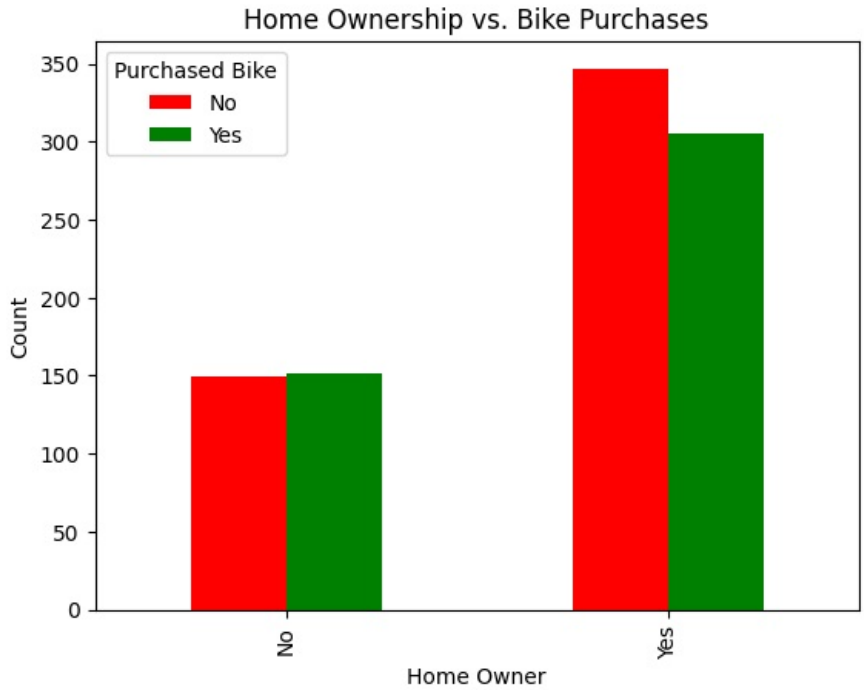


Does home ownership affect bike purchases?

In [273…

```
# Question 9: Home Ownership vs. Bike Purchases
home_bike = pd.crosstab(df['Home Owner'], df['Purchased Bike'])
plt.figure(figsize=(8, 5))
home_bike.plot(kind='bar', color=['red', 'green'])
plt.title('Home Ownership vs. Bike Purchases')
plt.xlabel('Home Owner')
plt.ylabel('Count')
plt.legend(title='Purchased Bike')
plt.show()
```

```
<Figure size 800x500 with 0 Axes>
```

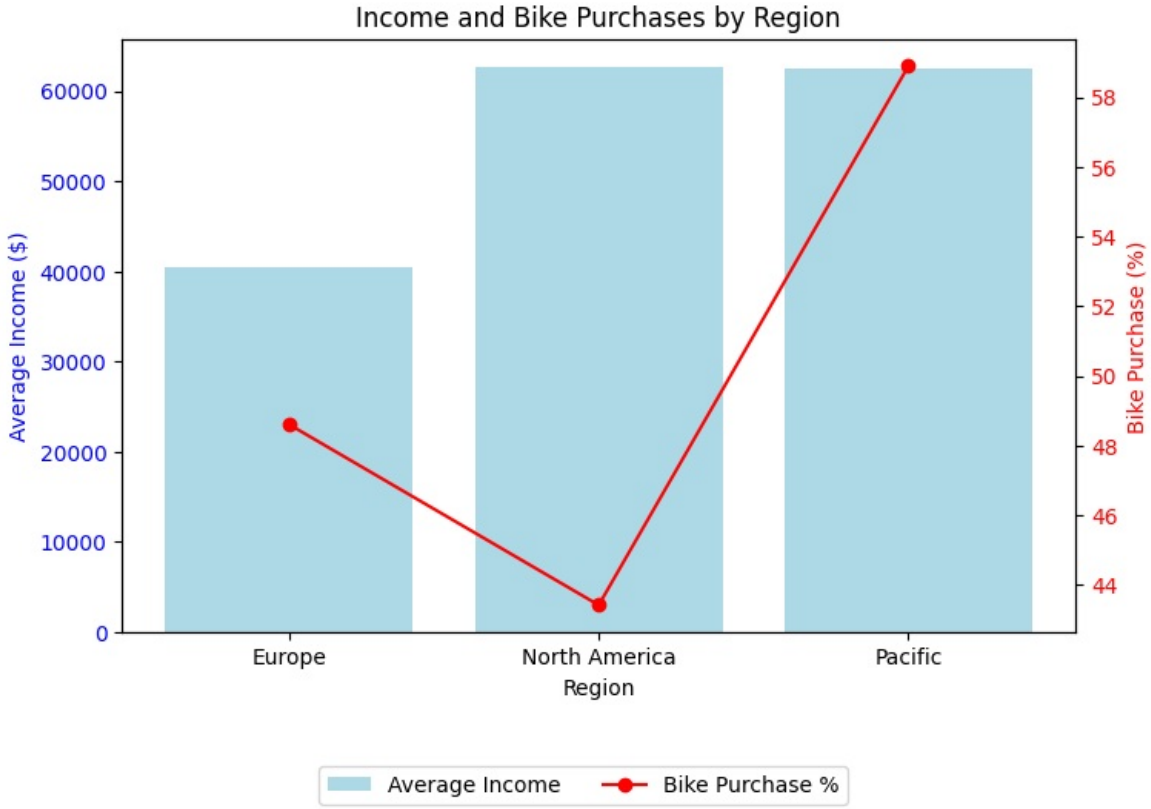


How does region affect income and bike purchases?

In [274…

```
# Question 10: Income and Bike Purchases by Region
income_region = df.groupby('Region')['Income'].mean()
bike_region = df.groupby('Region')['Purchased Bike'].apply(lambda x: (x == 'Yes').mean() * 100)
```

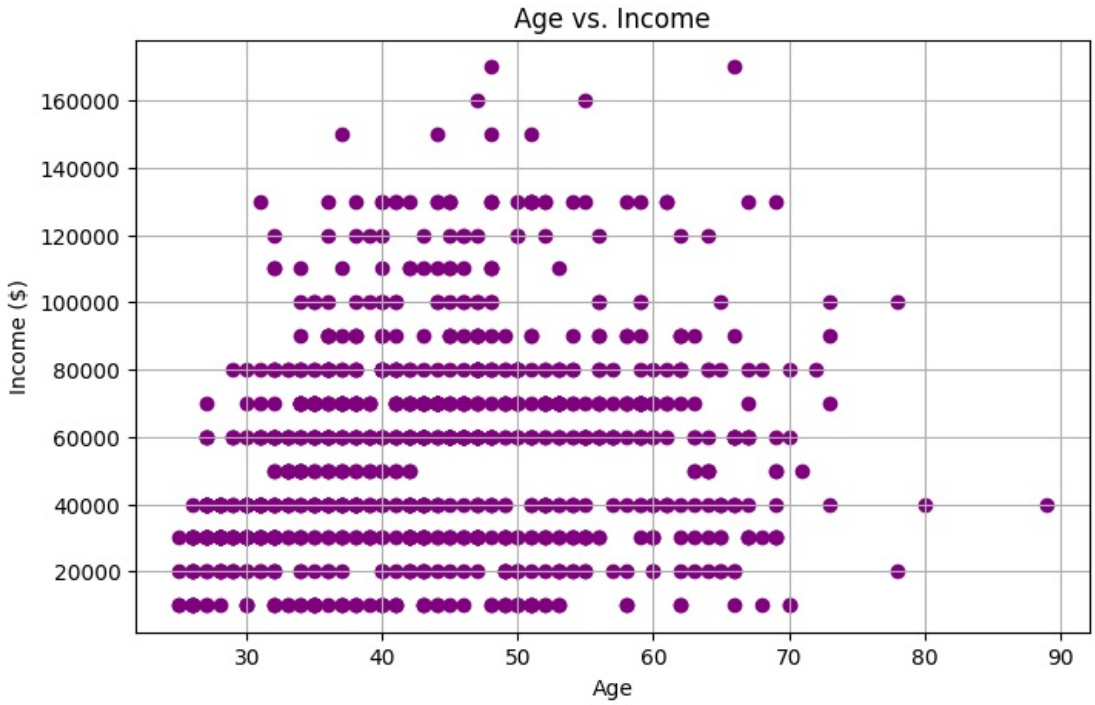
```
fig, ax1 = plt.subplots(figsize=(8, 5))
ax1.bar(income_region.index, income_region.values, color='lightblue', label='Average Income')
ax1.set_xlabel('Region')
ax1.set_ylabel('Average Income ($)', color='blue')
ax1.tick_params(axis='y', labelcolor='blue')

ax2 = ax1.twinx()
ax2.plot(bike_region.index, bike_region.values, color='red', marker='o', label='Bike Purchase %')
ax2.set_ylabel('Bike Purchase (%)', color='red')
ax2.tick_params(axis='y', labelcolor='red')

plt.title('Income and Bike Purchases by Region')
fig.legend(loc='upper center', bbox_to_anchor=(0.5, -0.05), ncol=2)
plt.show()
```



What is the relationship between age and income?

In [275…

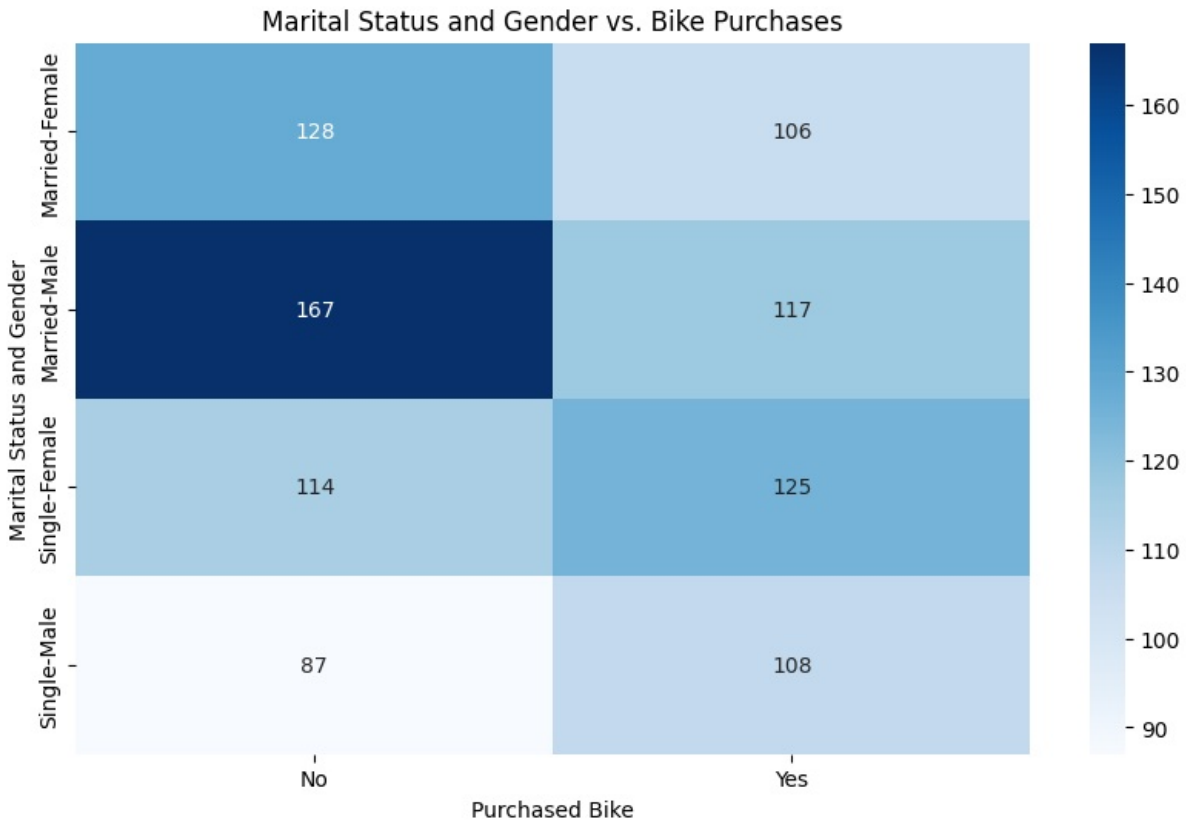
```
# Question 11: Age vs. Income
plt.figure(figsize=(8, 5))
plt.scatter(df['Age'], df['Income'], color='purple')
plt.title('Age vs. Income')
plt.xlabel('Age')
plt.ylabel('Income ($)')
plt.grid(True)
plt.show()
```



How does marital status and gender together affect bike purchases?

In [276…

```
# Question 12: Marital Status and Gender vs. Bike Purchases
marital_gender_bike = pd.crosstab([df['Marital Status'], df['Gender']], df['Purchased Bike'])
plt.figure(figsize=(10, 6))
sns.heatmap(marital_gender_bike, annot=True, cmap='Blues', fmt='d')
plt.title('Marital Status and Gender vs. Bike Purchases')
plt.xlabel('Purchased Bike')
plt.ylabel('Marital Status and Gender')
plt.show()
```

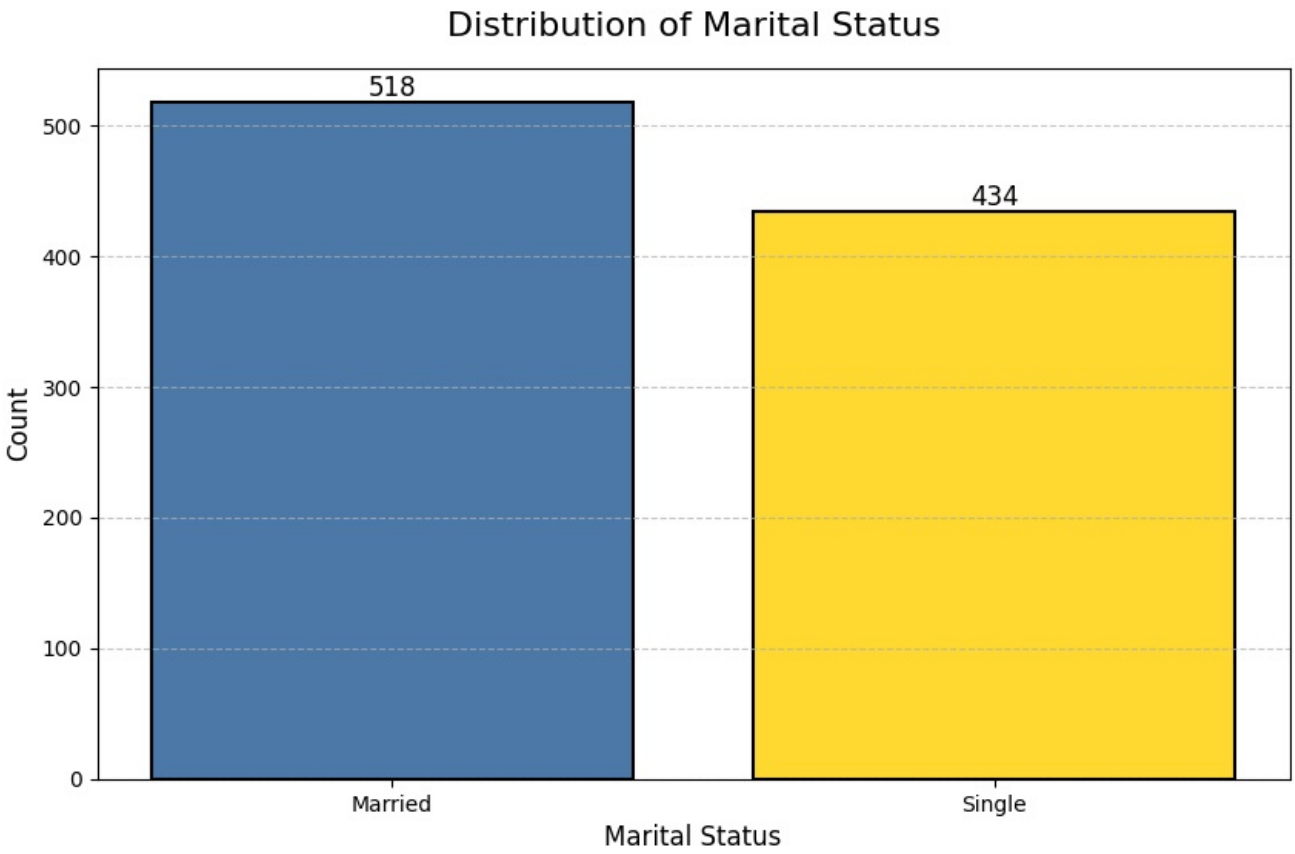


What is the distribution of Marital Status?

In [277…

```
marital_counts = df['Marital Status'].value_counts()
plt.figure(figsize=(10, 6))
bars = plt.bar(marital_counts.index, marital_counts.values, color=['#4C78A8', '#FFD92F'], edgecolor='black', li
plt.title('Distribution of Marital Status', fontsize=16, pad=15)
plt.xlabel('Marital Status', fontsize=12)
plt.ylabel('Count', fontsize=12)
for bar in bars:
    height = bar.get_height()
```

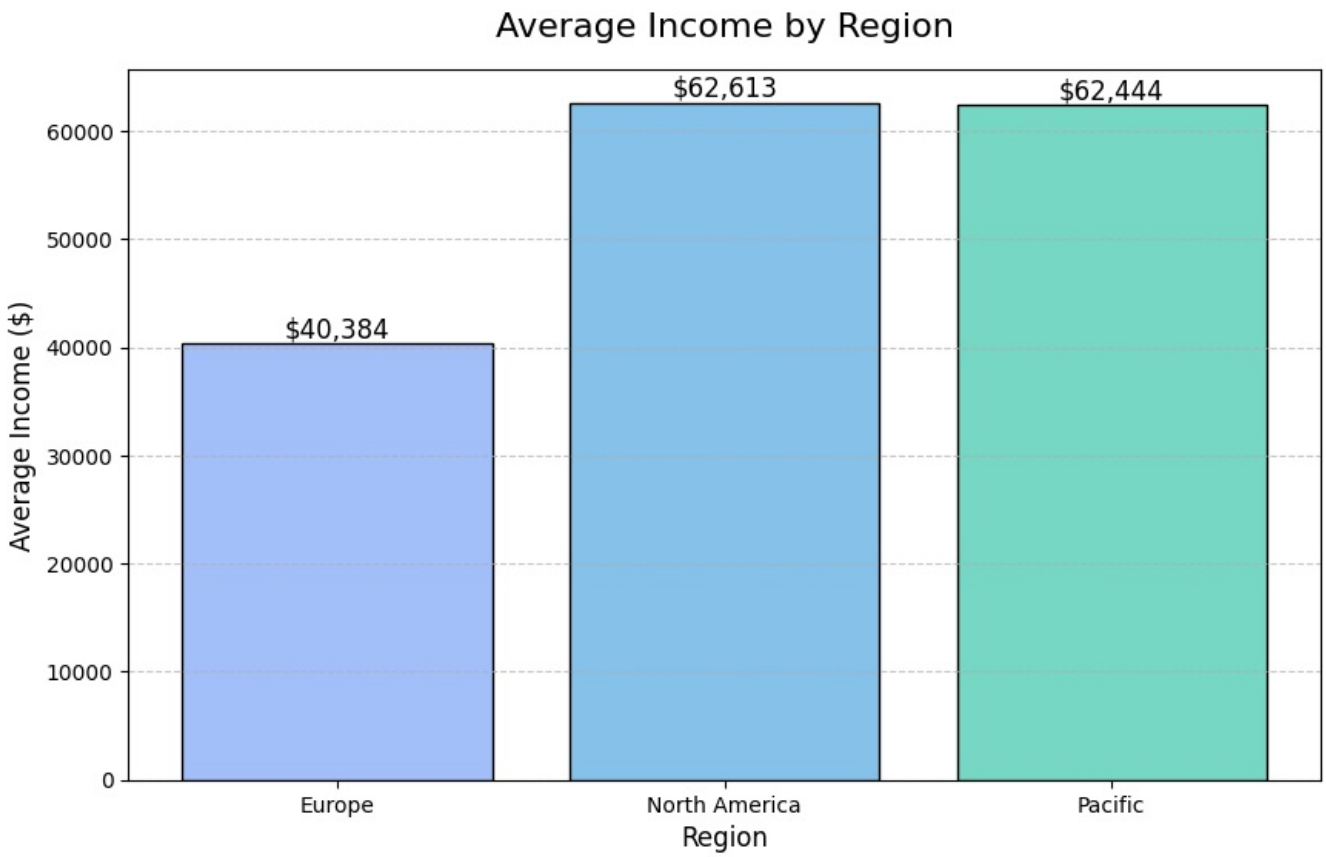
```
    plt.text(bar.get_x() + bar.get_width()/2., height, f'{int(height)}', ha='center', va='bottom', fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



What is the average income by region?

In [279…

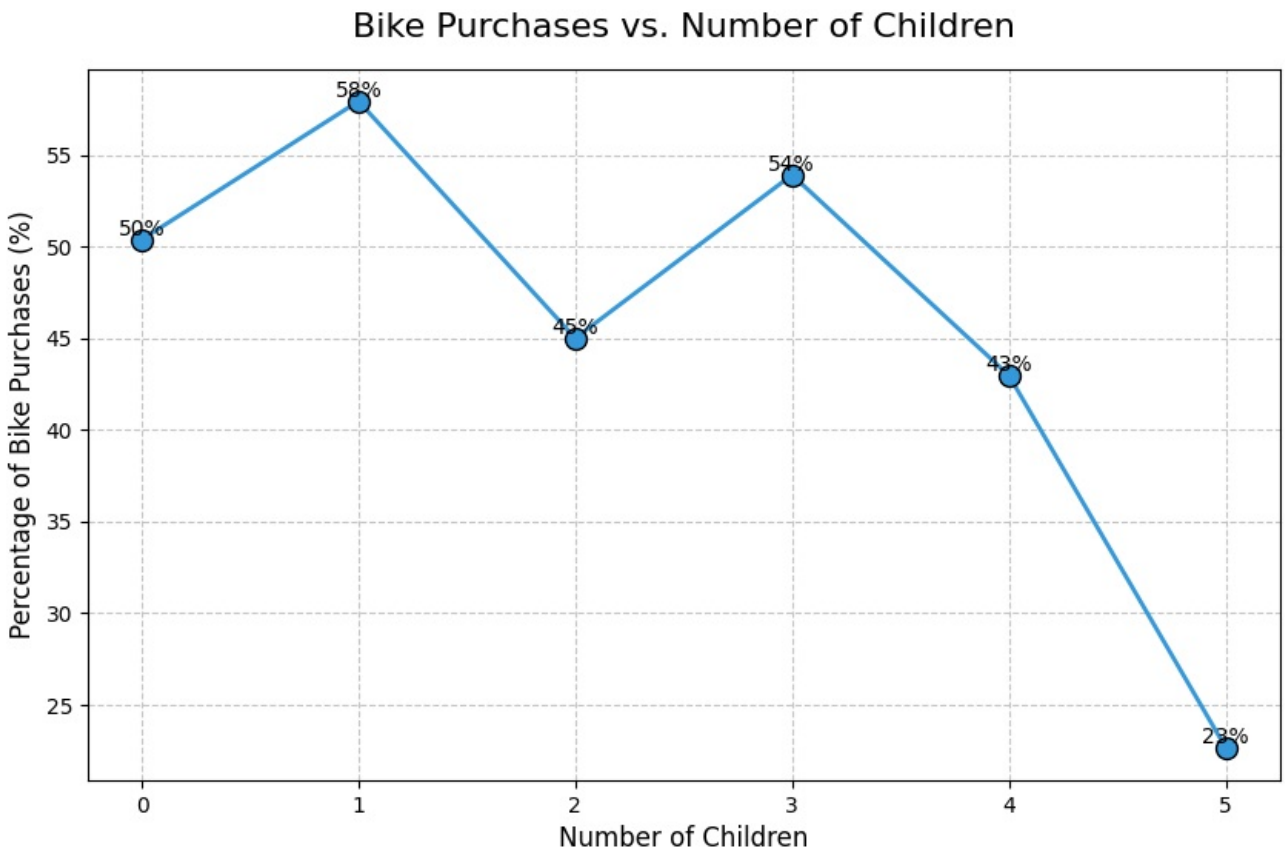
```
income_region = df.groupby('Region')['Income'].mean()
plt.figure(figsize=(10, 6))
bars = plt.bar(income_region.index, income_region.values, color=['#A3BFFA', '#85C1E9', '#76D7C4'], edgecolor='b
plt.title('Average Income by Region', fontsize=16, pad=15)
plt.xlabel('Region', fontsize=12)
plt.ylabel('Average Income ($)', fontsize=12)
for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2., bar.get_height(), f'${int(bar.get_height()):,}', ha='center', va
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



How does the number of children affect bike purchases?

In [280…

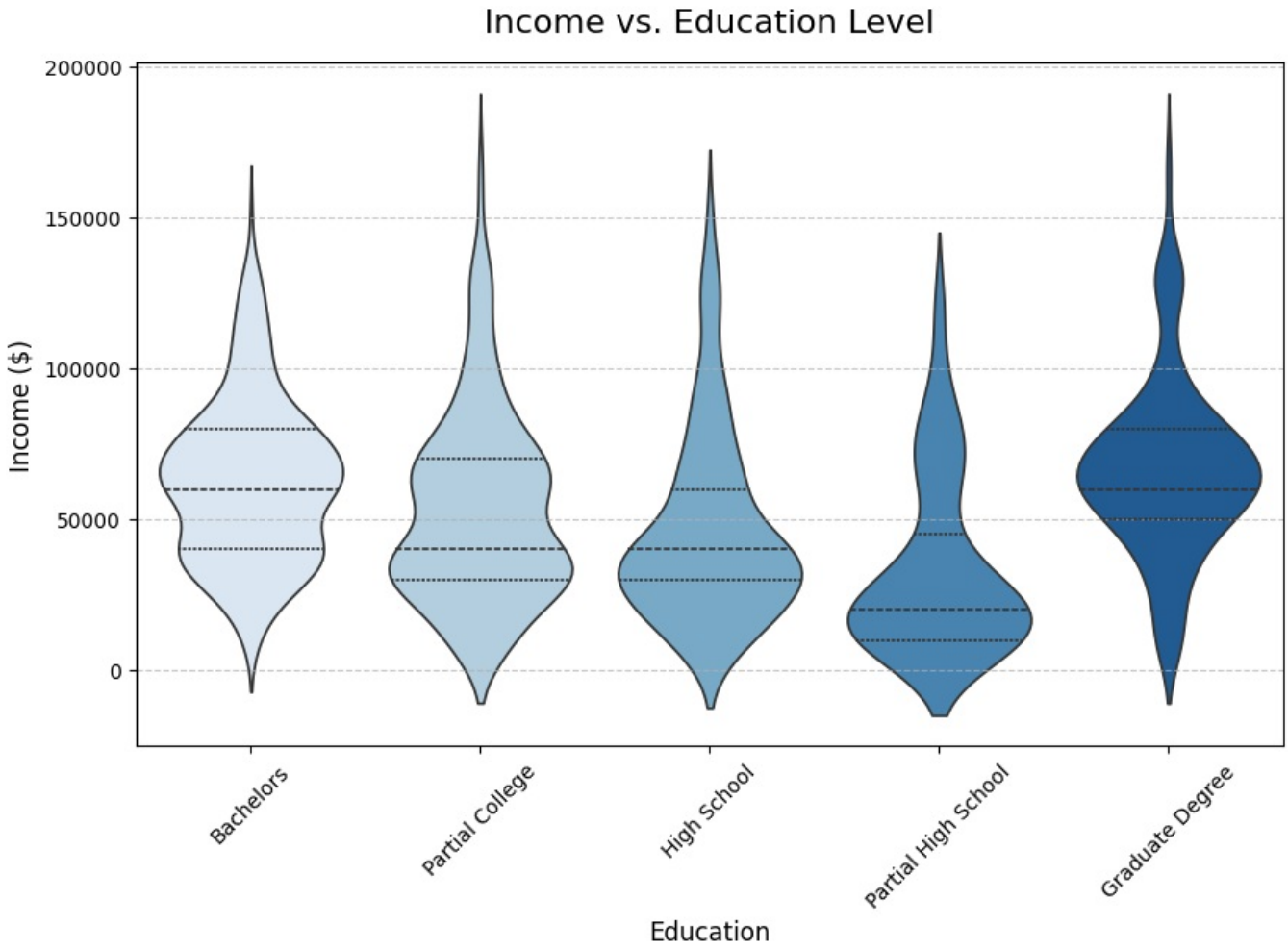
```
children_bike = df.groupby('Children')['Purchased Bike'].apply(lambda x: (x == 'Yes').mean() * 100)
plt.figure(figsize=(10, 6))
plt.plot(children_bike.index, children_bike.values, marker='o', color='#3498DB', linewidth=2, markersize=10, ma
plt.title('Bike Purchases vs. Number of Children', fontsize=16, pad=15)
plt.xlabel('Number of Children', fontsize=12)
plt.ylabel('Percentage of Bike Purchases (%)', fontsize=12)
for x, y in zip(children_bike.index, children_bike.values):
    plt.text(x, y, f'{y:.0f}%', ha='center', va='bottom', fontsize=10)
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



What is the relationship between education and income?

In [281…

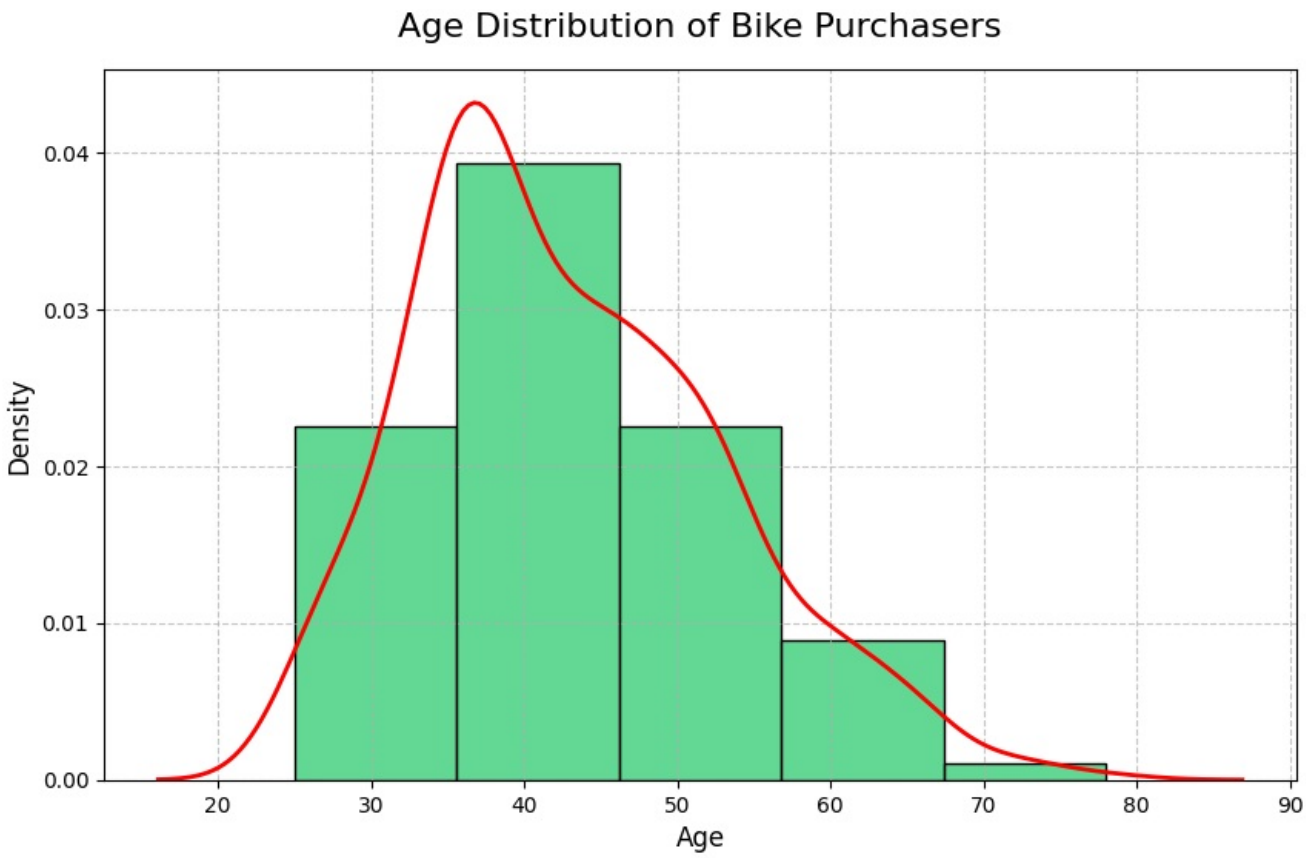
```
plt.figure(figsize=(10, 6))
sns.violinplot(x='Education', y='Income', data=df, palette='Blues', inner='quartile')
plt.title('Income vs. Education Level', fontsize=16, pad=15)
plt.xlabel('Education', fontsize=12)
plt.ylabel('Income ($)', fontsize=12)
plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



What is the age distribution of people who purchased bikes?

In [283…

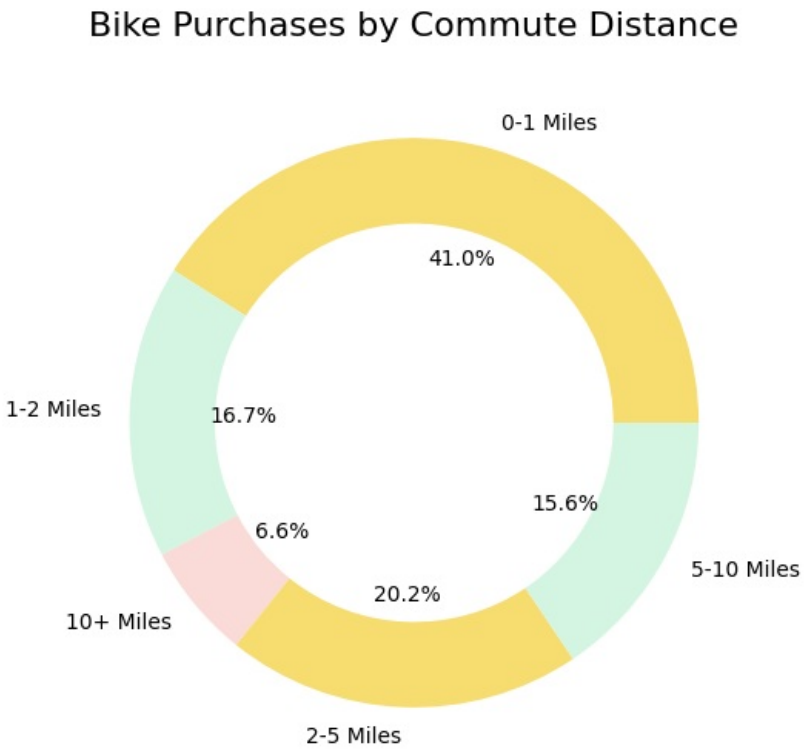
```
bike_ages = df[df['Purchased Bike'] == 'Yes']['Age']
plt.figure(figsize=(10, 6))
sns.histplot(bike_ages, bins=5, color='#2ECC71', edgecolor='black', stat='density')
sns.kdeplot(bike_ages, color='red', linewidth=2)
plt.title('Age Distribution of Bike Purchasers', fontsize=16, pad=15)
plt.xlabel('Age', fontsize=12)
plt.ylabel('Density', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



How does commute distance relate to bike purchases?

In [284…

```
commute_bike = df.groupby('Commute Distance')['Purchased Bike'].apply(lambda x: (x == 'Yes').sum())
plt.figure(figsize=(8, 6))
plt.pie(commute_bike, labels=commute_bike.index, autopct='%1.1f%%', colors=['#F7DC6F', '#D5F5E3', '#FADBD8'], w
plt.title('Bike Purchases by Commute Distance', fontsize=16, pad=15)
plt.show()
```

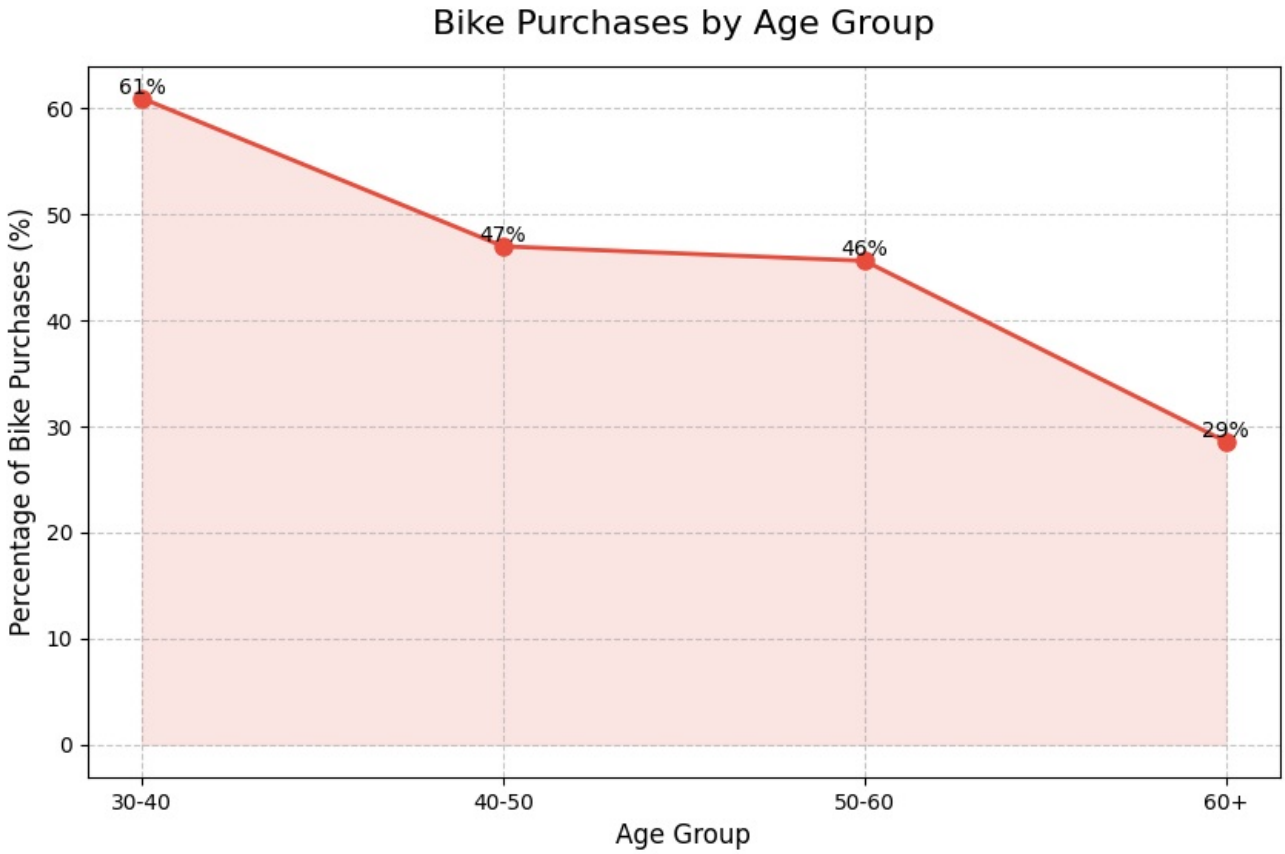


What is the trend of bike purchases by age group?

In [285…

```
df['Age Group'] = pd.cut(df['Age'], bins=[30, 40, 50, 60, 90], labels=['30-40', '40-50', '50-60', '60+'])
age_bike = df.groupby('Age Group')['Purchased Bike'].apply(lambda x: (x == 'Yes').mean() * 100)
plt.figure(figsize=(10, 6))
plt.fill_between(age_bike.index, age_bike.values, color='#FADBD8', alpha=0.7)
plt.plot(age_bike.index, age_bike.values, color='#E74C3C', linewidth=2, marker='o', markersize=8)
plt.title('Bike Purchases by Age Group', fontsize=16, pad=15)
plt.xlabel('Age Group', fontsize=12)
```

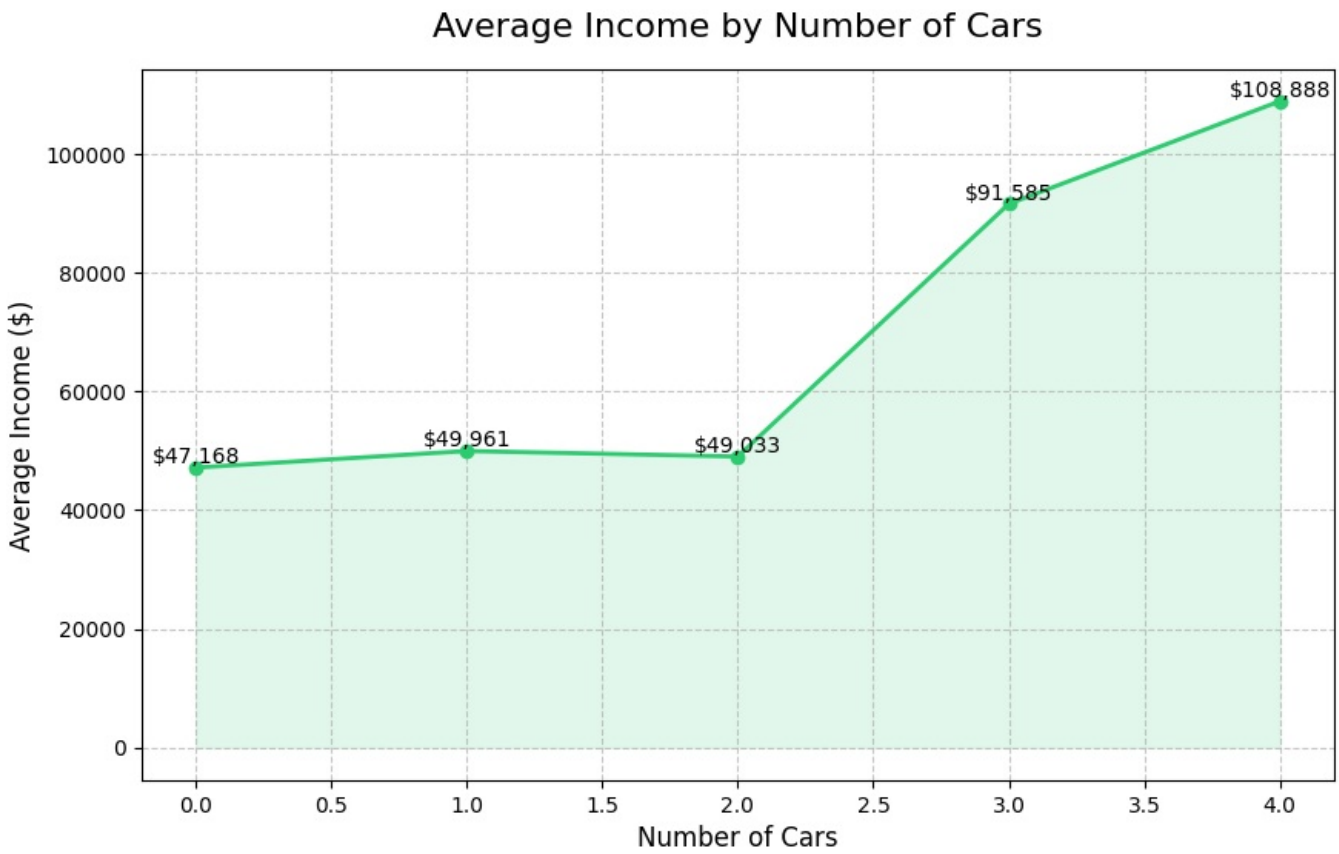
```
plt.ylabel('Percentage of Bike Purchases (%)', fontsize=12)
for x, y in age_bike.items():
    plt.text(x, y, f'{y:.0f}%', ha='center', va='bottom', fontsize=10)
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



How does the number of cars relate to income levels?

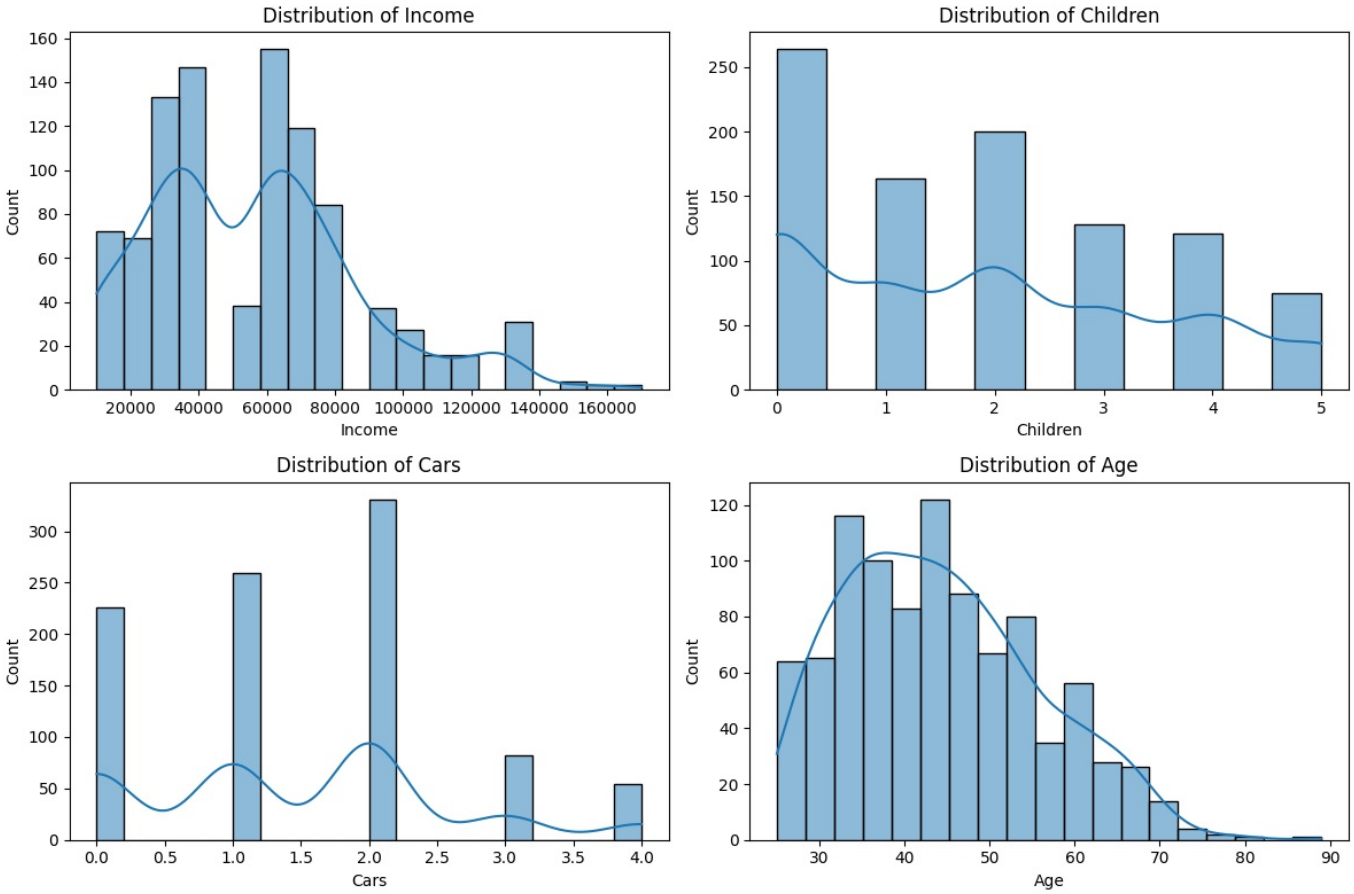
In [286…

```
cars_income = df.groupby('Cars')['Income'].mean()
plt.figure(figsize=(10, 6))
plt.fill_between(cars_income.index, cars_income.values, color='#D5F5E3', alpha=0.7)
plt.plot(cars_income.index, cars_income.values, color='#2ECC71', linewidth=2, marker='o')
plt.title('Average Income by Number of Cars', fontsize=16, pad=15)
plt.xlabel('Number of Cars', fontsize=12)
plt.ylabel('Average Income ($)', fontsize=12)
for x, y in cars_income.items():
    plt.text(x, y, f'${int(y):,}', ha='center', va='bottom', fontsize=10)
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



In [287…

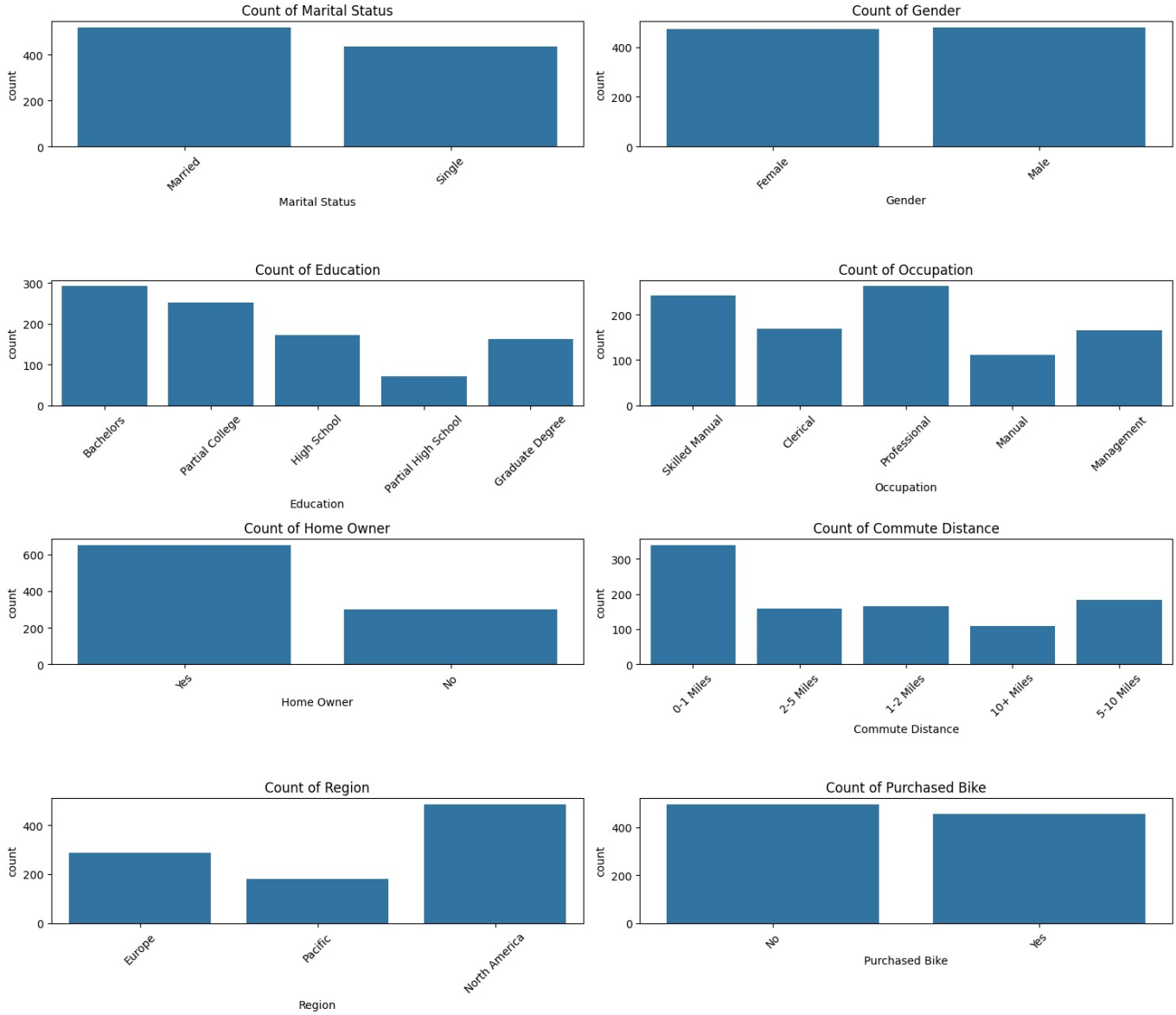
```
# 3. Visualizations
# Histogram for numerical variables
numerical_cols = ['Income', 'Children', 'Cars', 'Age']
plt.figure(figsize=(12, 8))
for i, col in enumerate(numerical_cols, 1):
    plt.subplot(2, 2, i)
    sns.histplot(df[col].dropna(), kde=True)
    plt.title(f'Distribution of {col}')
    plt.xlabel(col)
    plt.ylabel('Count')
plt.tight_layout()
plt.show()
```



In [288…

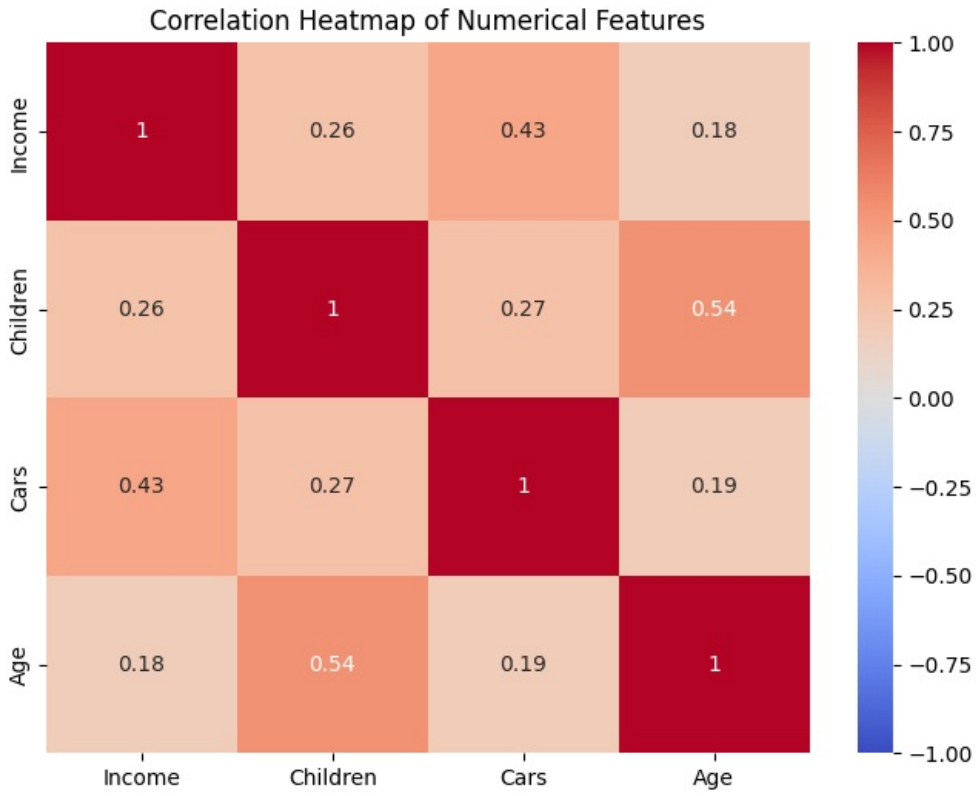
```
# Bar plots for categorical variables
```

```
categorical_cols = ['Marital Status', 'Gender', 'Education', 'Occupation',
                    'Home Owner', 'Commute Distance', 'Region', 'Purchased Bike']
plt.figure(figsize=(15, 13))
for i, col in enumerate(categorical_cols, 1):
    plt.subplot(4, 2, i)
    sns.countplot(data=df, x=col)
    plt.title(f'Count of {col}')
    plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



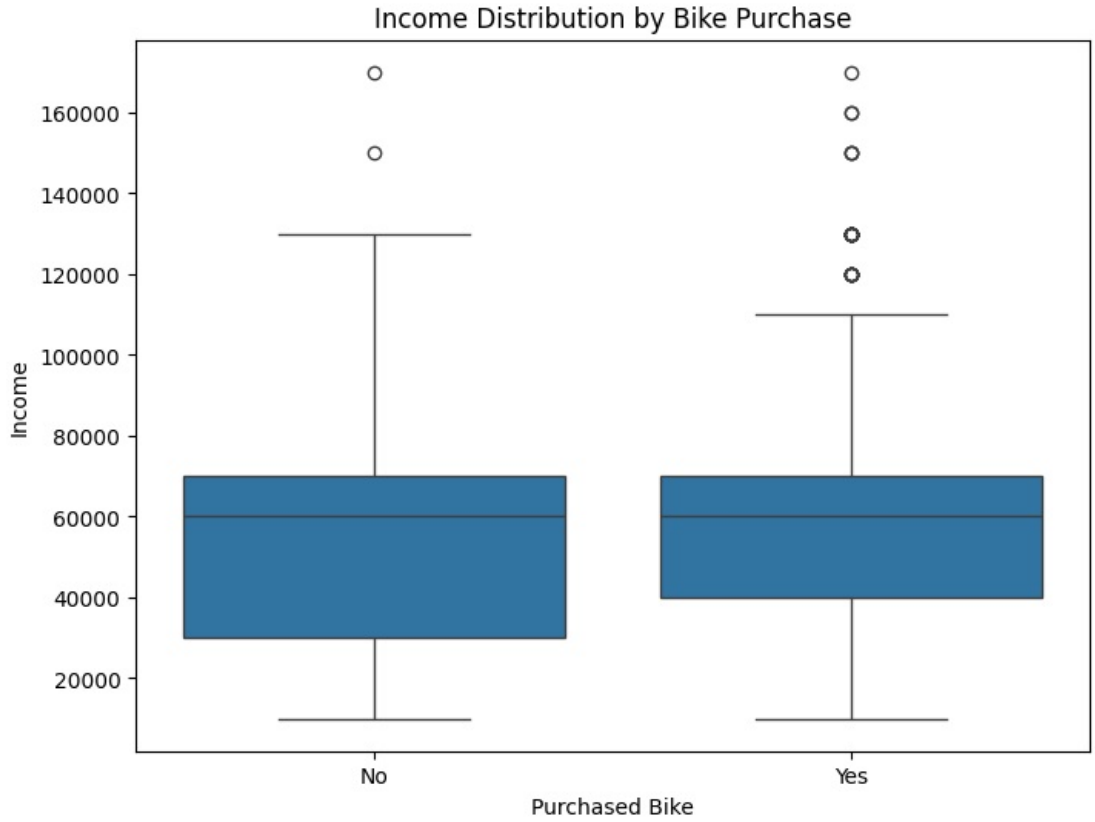
In [289…

```
# 4. Correlation Heatmap (for numerical variables)
plt.figure(figsize=(8, 6))
correlation_matrix = df[numerical_cols].corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title('Correlation Heatmap of Numerical Features')
plt.show()
```



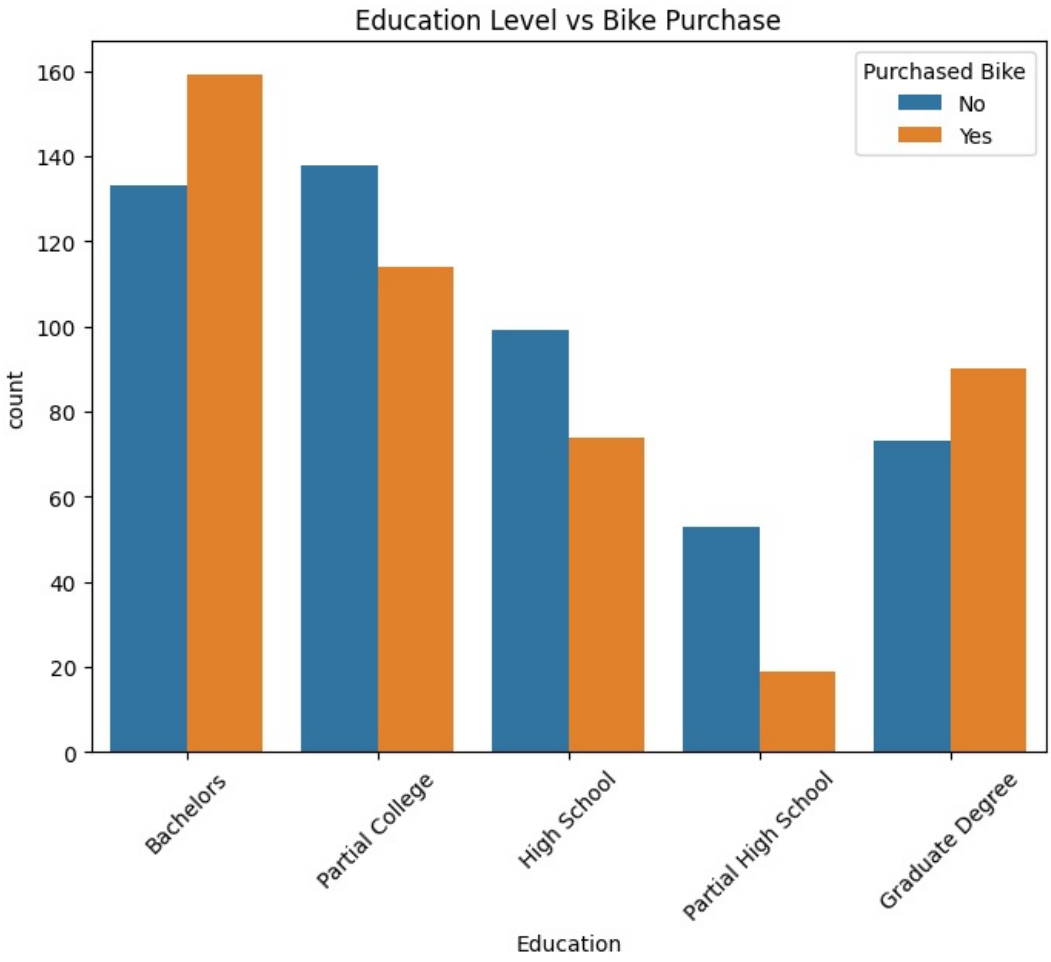
In [290…

```
# Box plot: Income vs Purchased Bike
plt.figure(figsize=(8, 6))
sns.boxplot(data=df, x='Purchased Bike', y='Income')
plt.title('Income Distribution by Bike Purchase')
plt.show()
```



In [291…

```
# Count plot: Education vs Purchased Bike
plt.figure(figsize=(8, 6))
sns.countplot(data=df, x='Education', hue='Purchased Bike')
plt.title('Education Level vs Bike Purchase')
plt.xticks(rotation=45)
plt.show()
```



In [292…

```
# 6. Trends, Patterns, and Potential Biases
print("\nEDA Observations:")
print("1. Distribution Analysis:")
for col in numerical_cols:
    skewness = stats.skew(df[col].dropna())
    print(f"- {col}: Skewness = {skewness:.2f} (positive skew indicates longer right tail)")

print("\n2. Categorical Patterns:")
print("- Check bar plots for imbalances (e.g., Gender, Region) that might indicate sampling bias.")
print("- Purchased Bike distribution: Check if 'Yes'/'No' is balanced.")

print("\n3. Correlations:")
for i in range(len(numerical_cols)):
    for j in range(i+1, len(numerical_cols)):
        corr = correlation_matrix.iloc[i, j]
        if abs(corr) > 0.5:
            print(f"- Strong correlation between {numerical_cols[i]} and {numerical_cols[j]}: {corr:.2f}")

print("\n4. Potential Biases:")
print("- Missing Gender values (e.g., NaN in row 3) could bias gender-related insights.")
print("- Regional distribution might overrepresent Europe, potentially skewing results.")
print("- Income outliers (visible in box plot) could affect regression models.")
```

```
EDA Observations:
1. Distribution Analysis:
- Income: Skewness = 0.72 (positive skew indicates longer right tail)
- Children: Skewness = 0.40 (positive skew indicates longer right tail)
- Cars: Skewness = 0.41 (positive skew indicates longer right tail)
- Age: Skewness = 0.50 (positive skew indicates longer right tail)

2. Categorical Patterns:
- Check bar plots for imbalances (e.g., Gender, Region) that might indicate sampling bias.
- Purchased Bike distribution: Check if 'Yes'/'No' is balanced.

3. Correlations:
- Strong correlation between Children and Age: 0.54

4. Potential Biases:
- Missing Gender values (e.g., NaN in row 3) could bias gender-related insights.
- Regional distribution might overrepresent Europe, potentially skewing results.
- Income outliers (visible in box plot) could affect regression models.
```

Model Development & Evaluation

Linear Regression

In [293…
```
X = dfd.drop(columns=['ID', 'Income'])  # All preprocessed variables except 'Income'
y = dfd['Income']  # Target variable is 'Income'
```

In [294…
```
# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

In [295…
```
# Initialize and train the Linear Regression model
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)
```

Out[295]:
```
▾ LinearRegression
LinearRegression()
```

In [296…
```
# Make predictions
y_pred = lr_model.predict(X_test)
```

In [297…
```
# Evaluate the model
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

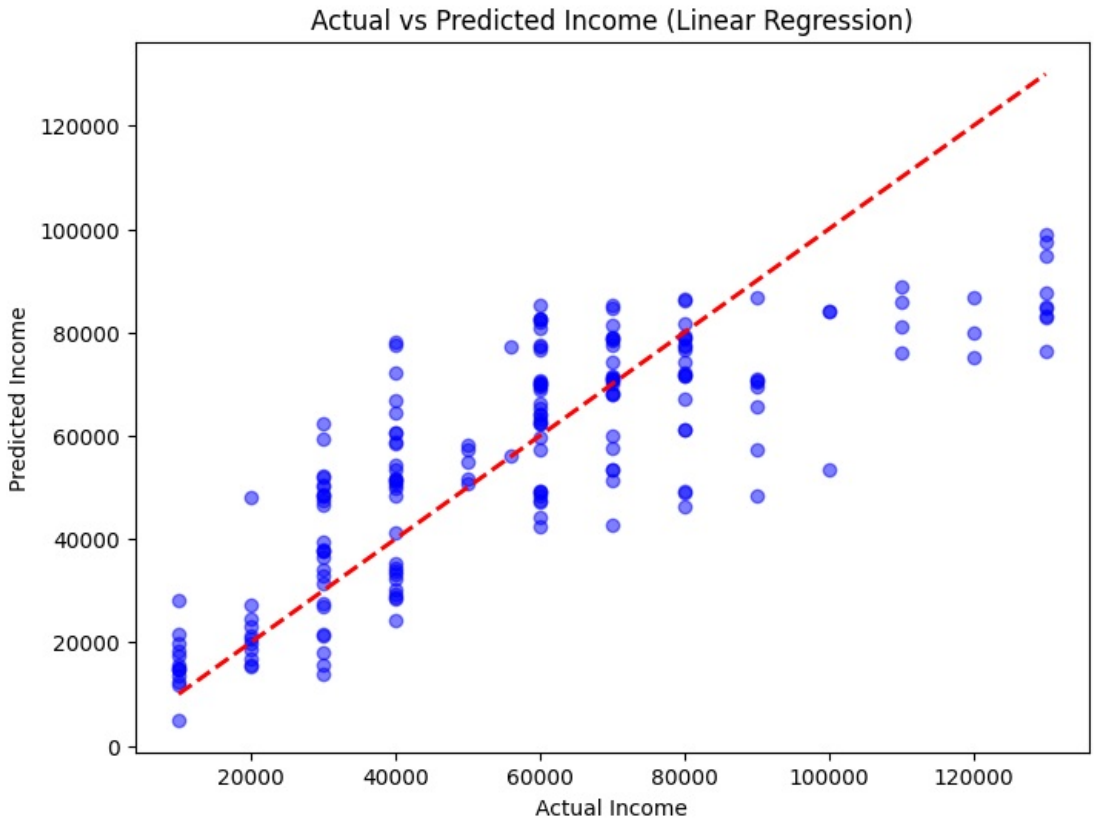
In [298…
```
# Print evaluation metrics
print("Linear Regression Performance:")
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"R-squared (R2): {r2:.2f}")
```

```
Linear Regression Performance:
Mean Squared Error (MSE): 328959539.71
R-squared (R2): 0.63
```

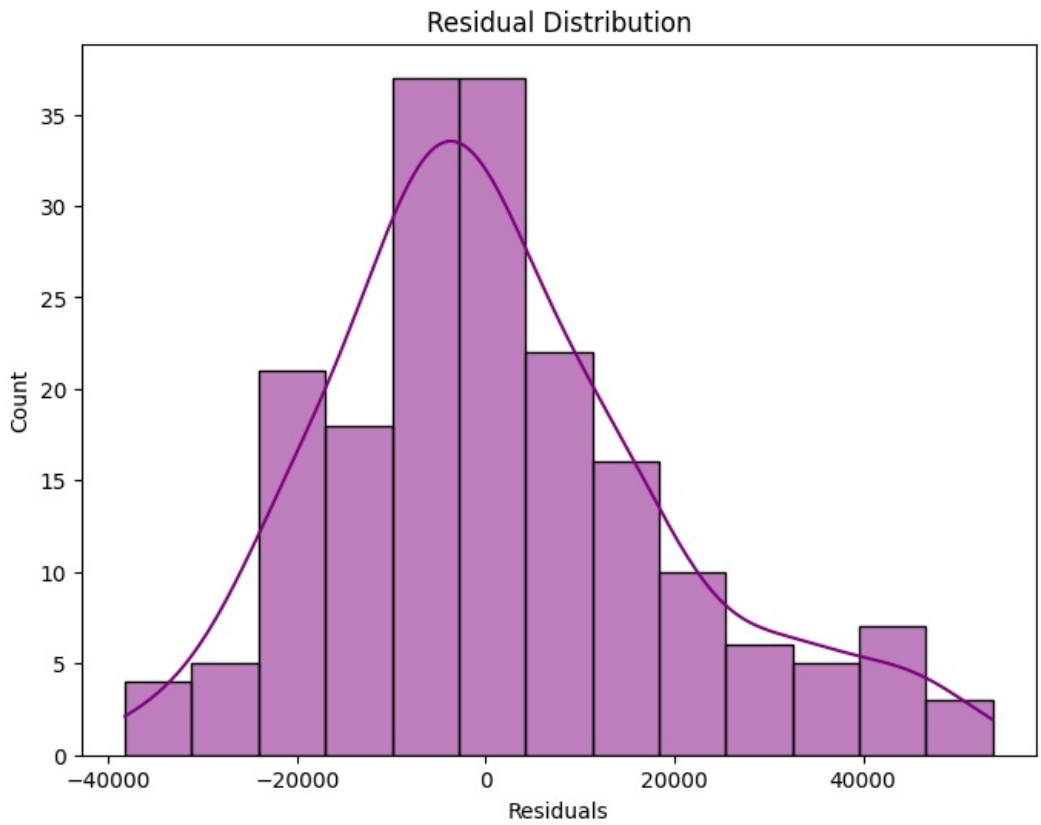
In [299…
```
# Display coefficients
coefficients = pd.DataFrame(lr_model.coef_, X.columns, columns=['Coefficient'])
print("\nModel Coefficients:")
print(coefficients)
```

```
Model Coefficients:
                              Coefficient
Marital Status                 -315.601727
Gender                        -2193.324903
Children                        984.804744
Education                      2496.009724
Home Owner                     -582.618286
Cars                           5491.815800
Age                            -372.760418
Purchased Bike                 2192.613832
Occupation_Management         51513.679427
Occupation_Manual            -17802.979087
Occupation_Professional       39727.661271
Occupation_Skilled Manual     21564.174201
Region_North America          -6431.939981
Region_Pacific                -6070.140359
Commute Distance_1-2 Miles     1568.096824
Commute Distance_10+ Miles    -4117.262246
Commute Distance_2-5 Miles      533.359732
Commute Distance_5-10 Miles   -2181.717402
```

In [300…
```
# Visualize Actual vs Predicted values
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, color='blue', alpha=0.5)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2)
plt.xlabel('Actual Income')
plt.ylabel('Predicted Income')
plt.title('Actual vs Predicted Income (Linear Regression)')
plt.show()
```



In [301…
```
# Residual plot
residuals = y_test - y_pred
plt.figure(figsize=(8, 6))
sns.histplot(residuals, kde=True, color='purple')
plt.xlabel('Residuals')
plt.title('Residual Distribution')
plt.show()
```



Classification

In [302…
```
X = dfd.drop(['ID'], axis=1)  # Features for classification
y_class = df['Purchased Bike']
```

In [303…
```
X_train, X_test, y_train_class, y_test_class = train_test_split(X, y_class, test_size=0.2, random_state=42)
```

In [304…
```
print(X_train.shape)
```

```
print(X_test.shape)
print(y_train_class.shape)
print(y_test_class.shape)
```

```
(761, 19)
(191, 19)
(761,)
(191,)
```

In [305…

```
# Train Logistic Regression
log_model = LogisticRegression(max_iter=1000, random_state=42)  # Increased max_iter for convergence
log_model.fit(X_train, y_train_class)
```

Out[305]:

```
LogisticRegression
LogisticRegression(max_iter=1000, random_state=42)
```

In [306…

```
# Make predictions
log_pred = log_model.predict(X_test)
```

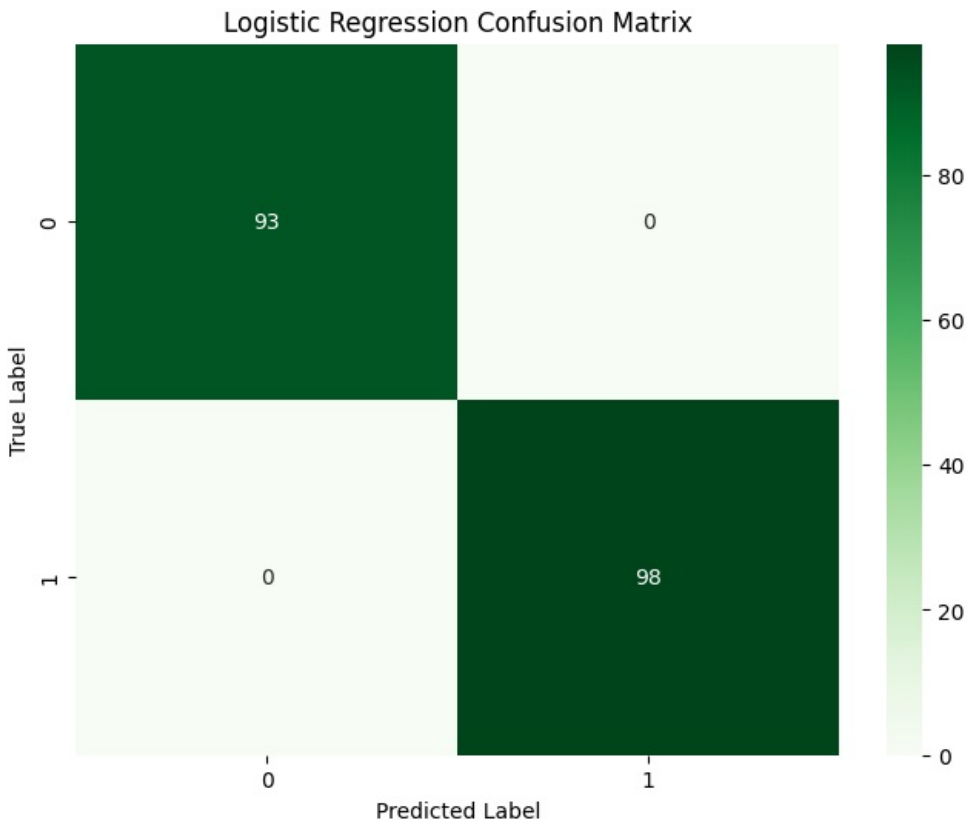
In [307…

```
# Evaluate Logistic Regression
log_accuracy = accuracy_score(y_test_class, log_pred)
print("Logistic Regression Results (Predicting Purchased Bike):")
print(f"Accuracy: {log_accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_test_class, log_pred))
```

```
Logistic Regression Results (Predicting Purchased Bike):
Accuracy: 1.0000

Classification Report:
              precision    recall  f1-score   support

          No       1.00      1.00      1.00        93
         Yes       1.00      1.00      1.00        98

    accuracy                           1.00       191
   macro avg       1.00      1.00      1.00       191
weighted avg       1.00      1.00      1.00       191
```

In [308…

```
# Confusion Matrix Visualization for Logistic Regression
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test_class, log_pred), annot=True, fmt='d', cmap='Greens')
plt.title('Logistic Regression Confusion Matrix')
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.show()
```



Decision Tree

In [309…
```python
# Train Decision Tree
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train_class)
```

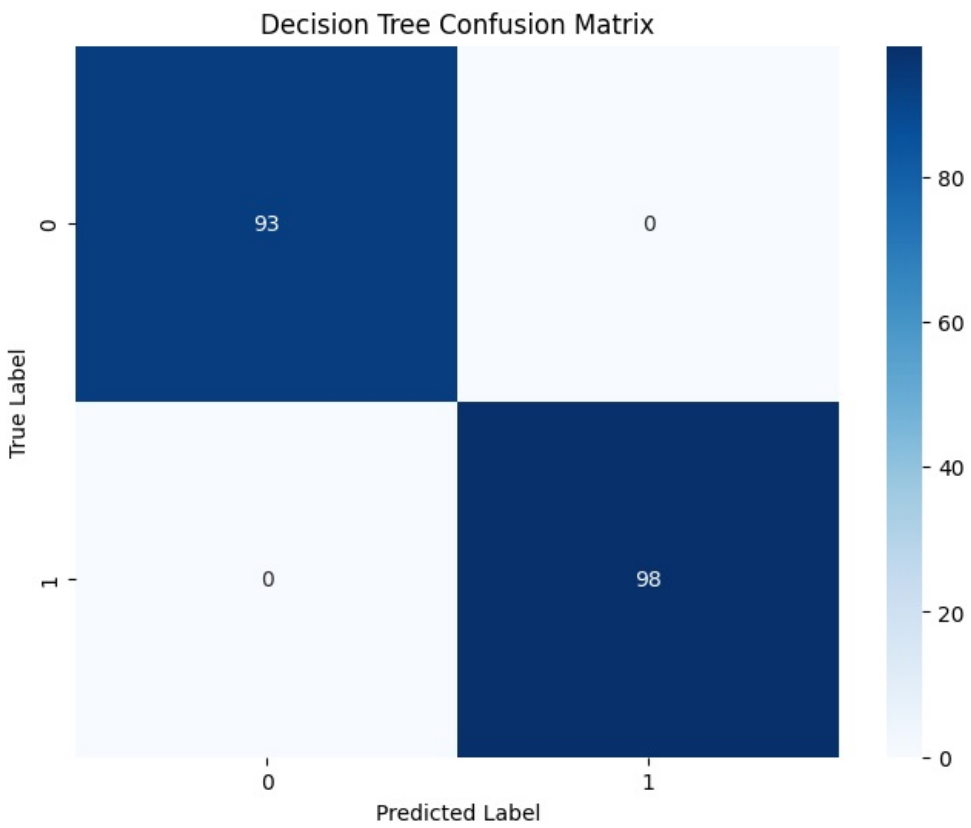
Out[309]:
```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

In [310…
```python
# Make predictions
dt_pred = dt_model.predict(X_test)
```

In [311…
```python
# Evaluate Decision Tree
dt_accuracy = accuracy_score(y_test_class, dt_pred)
print("\nDecision Tree Classifier Results (Predicting Purchased Bike):")
print(f"Accuracy: {dt_accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_test_class, dt_pred))
```

```
Decision Tree Classifier Results (Predicting Purchased Bike):
Accuracy: 1.0000

Classification Report:
              precision    recall  f1-score   support

          No       1.00      1.00      1.00        93
         Yes       1.00      1.00      1.00        98

    accuracy                           1.00       191
   macro avg       1.00      1.00      1.00       191
weighted avg       1.00      1.00      1.00       191
```

In [312…
```python
# Confusion Matrix Visualization for Decision Tree
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test_class, dt_pred), annot=True, fmt='d', cmap='Blues')
plt.title('Decision Tree Confusion Matrix')
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.show()
```



Feature Selection

Regression

In [313…
```python
Xl = dfd.drop(columns=['ID', 'Income'])  # All preprocessed variables except 'Income'
yl = dfd['Income']  # Target variable is 'Income'
```

In [314…

```python
# Perform feature selection using SelectKBest with f_regression
# Select the top 5 features (you can adjust k as needed)
k = 12  # Number of top features to select
selector = SelectKBest(score_func=f_regression, k=k)
X_selected = selector.fit_transform(Xl, yl)
```

In [315…

```python
# Get the selected feature names
selected_mask = selector.get_support()  # Boolean mask of selected features
selected_features = Xl.columns[selected_mask].tolist()
print("Selected Features:", selected_features)
```

```
Selected Features: ['Marital Status', 'Children', 'Education', 'Cars', 'Age', 'Occupation_Management', 'Occupati
on_Manual', 'Occupation_Professional', 'Region_North America', 'Region_Pacific', 'Commute Distance_10+ Miles', '
Commute Distance_2-5 Miles']
```

In [316…

```python
# Create a new DataFrame with only the selected features
Xl = Xl[selected_features]

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(Xl, yl, test_size=0.2, random_state=42)
```

In [317…

```python
# Initialize and train the Linear Regression model
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)
```

Out[317]:

```
▾ LinearRegression
LinearRegression()
```

In [318…

```python
# Make predictions
y_pred = lr_model.predict(X_test)

# Evaluate the model
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

In [319…

```python
# Print evaluation metrics
print("\nLinear Regression Performance (with Feature Selection):")
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"R-squared (R2): {r2:.2f}")
```

```
Linear Regression Performance (with Feature Selection):
Mean Squared Error (MSE): 360837285.77
R-squared (R2): 0.60
```

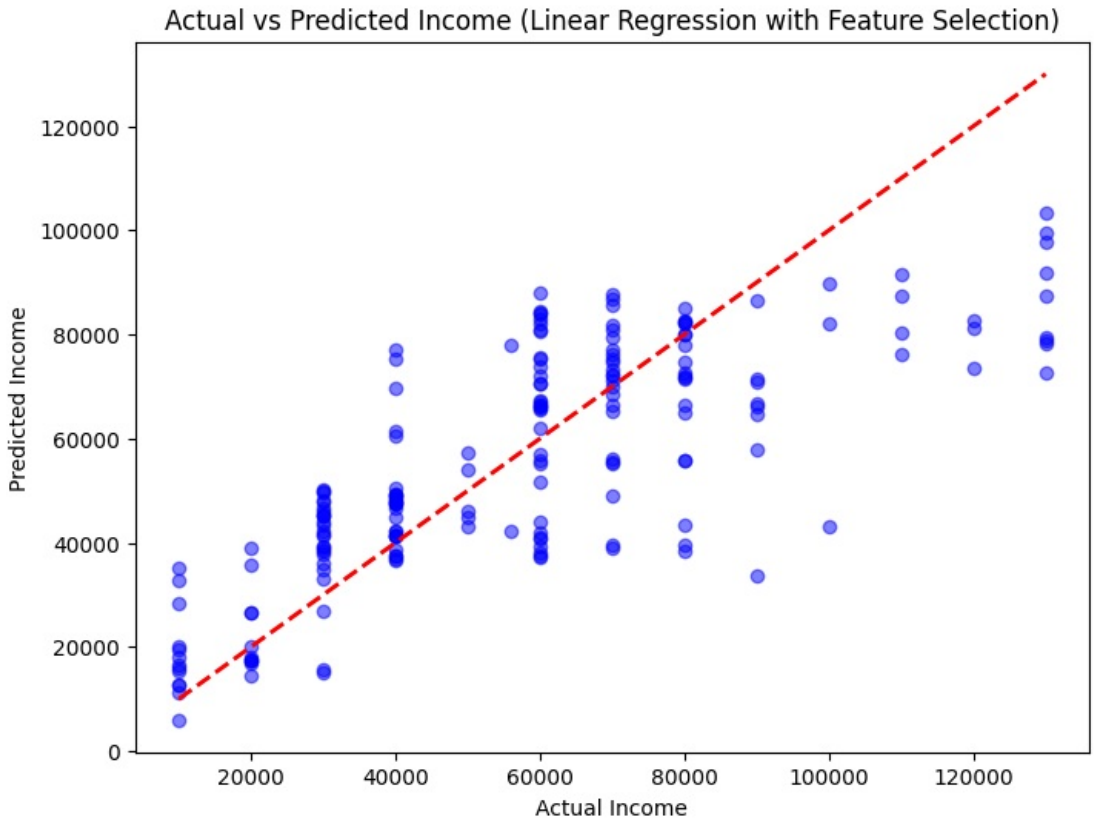
In [320…

```python
# Display coefficients
coefficients = pd.DataFrame(lr_model.coef_, Xl.columns, columns=['Coefficient'])
print("\nModel Coefficients:")
print(coefficients)
```

```
Model Coefficients:
                            Coefficient
Marital Status               -363.181440
Children                     1334.097390
Education                    3562.255737
Cars                         6318.715188
Age                          -434.140389
Occupation_Management       35633.251506
Occupation_Manual          -24774.643744
Occupation_Professional     24817.544555
Region_North America         1339.264715
Region_Pacific               -817.380399
Commute Distance_10+ Miles  -3258.956599
Commute Distance_2-5 Miles   1264.472263
```

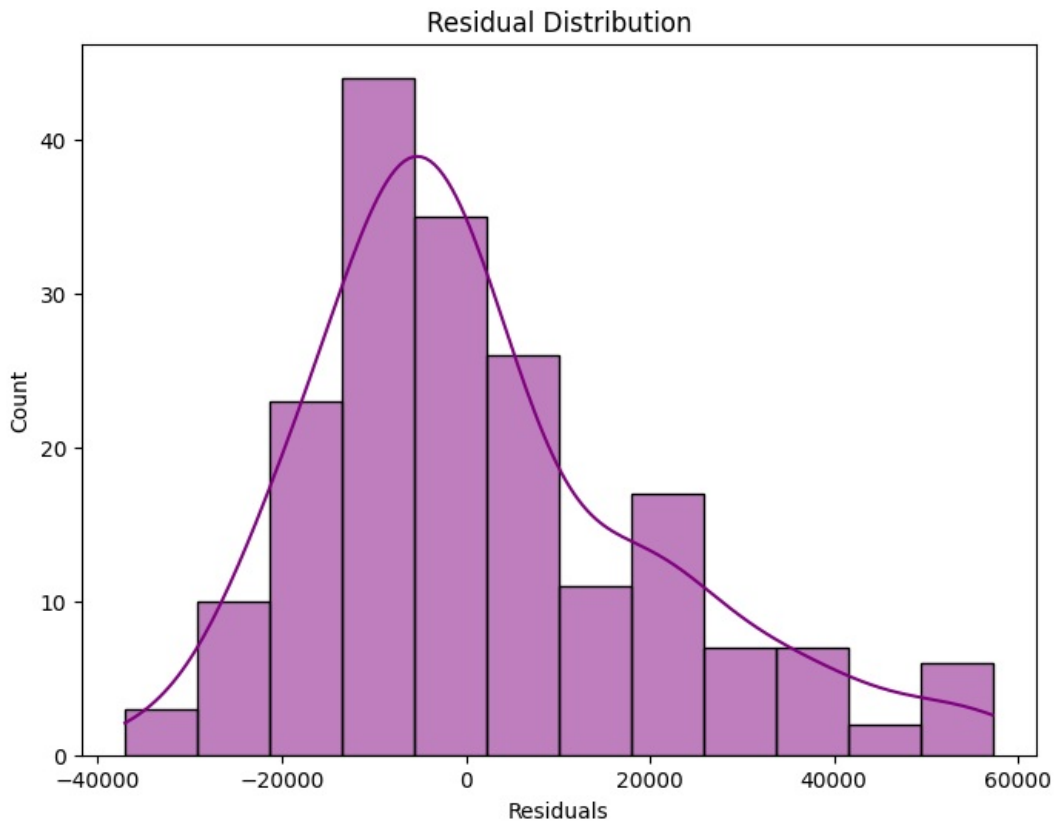
In [321…

```python
# Visualize Actual vs Predicted values
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, color='blue', alpha=0.5)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2)
plt.xlabel('Actual Income')
plt.ylabel('Predicted Income')
plt.title('Actual vs Predicted Income (Linear Regression with Feature Selection)')
plt.show()
```



In [322…

```
# Residual plot
residuals = y_test - y_pred
plt.figure(figsize=(8, 6))
sns.histplot(residuals, kde=True, color='purple')
plt.xlabel('Residuals')
plt.title('Residual Distribution')
plt.show()
```



Classification

In [323…

```
# Apply SelectKBest with chi-squared test to select top k features
k = 10  # Select top 10 features (you can adjust this number)
selector = SelectKBest(score_func=chi2, k=k)
X_selected = selector.fit_transform(X, y_class)
```

In [324…

```
# Get selected feature names
```

```
selected_features = X.columns[selector.get_support()].tolist()
print("Selected Features:", selected_features)
```

```
Selected Features: ['Income', 'Children', 'Education', 'Cars', 'Age', 'Purchased Bike', 'Region_Pacific', 'Commu
te Distance_10+ Miles', 'Commute Distance_2-5 Miles', 'Commute Distance_5-10 Miles']
```

In [325…

```
# Update X with selected features
X = pd.DataFrame(X_selected, columns=selected_features)
```

In [326…

```
X_train, X_test, y_train_class, y_test_class = train_test_split(X, y_class, test_size=0.2, random_state=42)
```

In [327…

```
# Train Logistic Regression
log_model = LogisticRegression(max_iter=1000, random_state=42)  # Increased max_iter for convergence
log_model.fit(X_train, y_train_class)
```

Out[327]:

```
▾ LogisticRegression
LogisticRegression(max_iter=1000, random_state=42)
```

In [328…

```
# Make predictions
log_pred = log_model.predict(X_test)
```

In [329…

```
log_prob = log_model.predict_proba(X_test)[:, 1]  # Probabilities for ROC curve
```

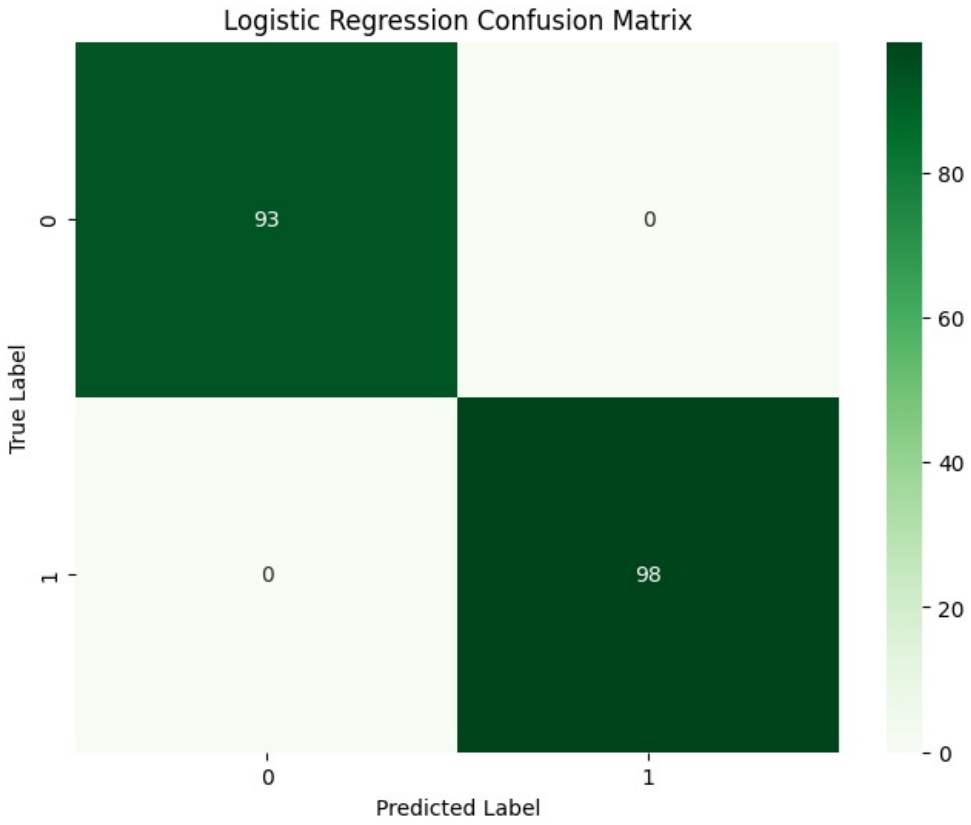
In [330…

```
# Evaluate Logistic Regression
log_accuracy = accuracy_score(y_test_class, log_pred)
print("Logistic Regression Results (Predicting Purchased Bike):")
print(f"Accuracy: {log_accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_test_class, log_pred))
```

```
Logistic Regression Results (Predicting Purchased Bike):
Accuracy: 1.0000

Classification Report:
              precision    recall  f1-score   support

          No       1.00      1.00      1.00        93
         Yes       1.00      1.00      1.00        98

    accuracy                           1.00       191
   macro avg       1.00      1.00      1.00       191
weighted avg       1.00      1.00      1.00       191
```

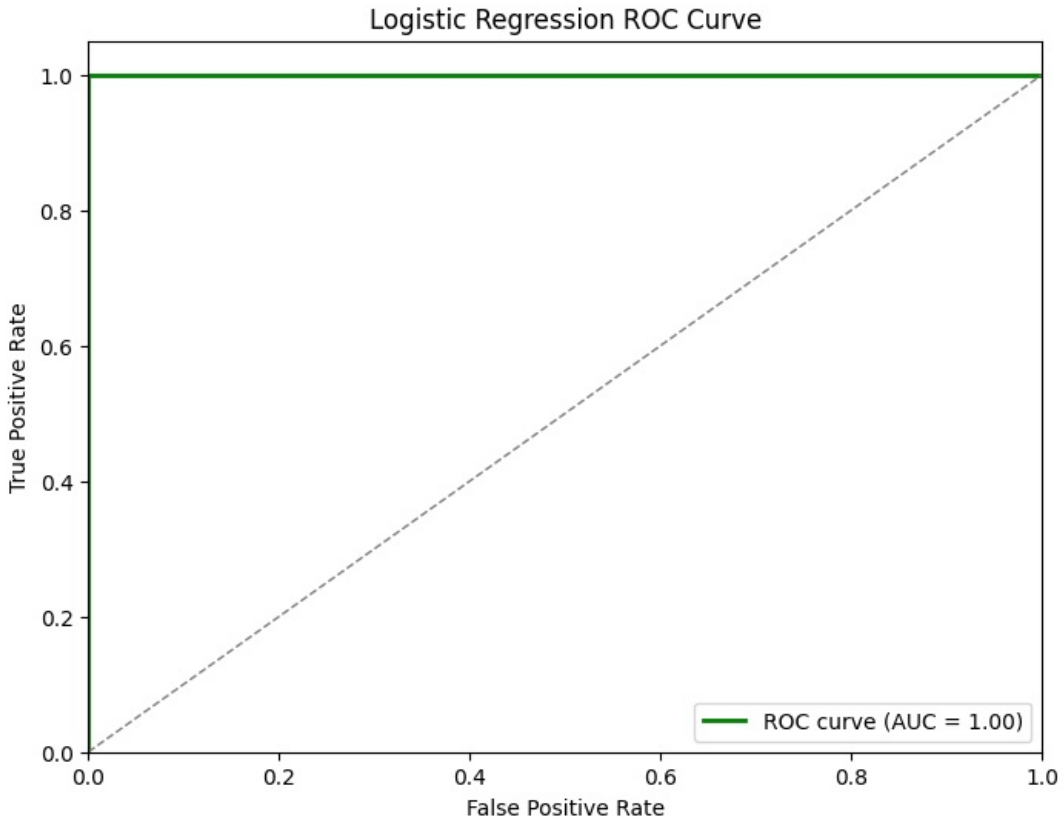
In [331…

```
# Confusion Matrix Visualization for Logistic Regression
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test_class, log_pred), annot=True, fmt='d', cmap='Greens')
plt.title('Logistic Regression Confusion Matrix')
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.show()
```



In [332…

```
# ROC Curve for Logistic Regression
fpr_log, tpr_log, _ = roc_curve(y_test_class.map({'No': 0, 'Yes': 1}), log_prob)
roc_auc_log = auc(fpr_log, tpr_log)
plt.figure(figsize=(8, 6))
plt.plot(fpr_log, tpr_log, color='green', lw=2, label=f'ROC curve (AUC = {roc_auc_log:.2f})')
plt.plot([0, 1], [0, 1], color='gray', lw=1, linestyle='--')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Logistic Regression ROC Curve')
plt.legend(loc="lower right")
plt.show()
```



In [333…

```
# Train Decision Tree
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train_class)
```

Out[333]:

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

In [334…

```python
# Make predictions
dt_pred = dt_model.predict(X_test)
```

In [335…

```python
dt_prob = dt_model.predict_proba(X_test)[:, 1]
```

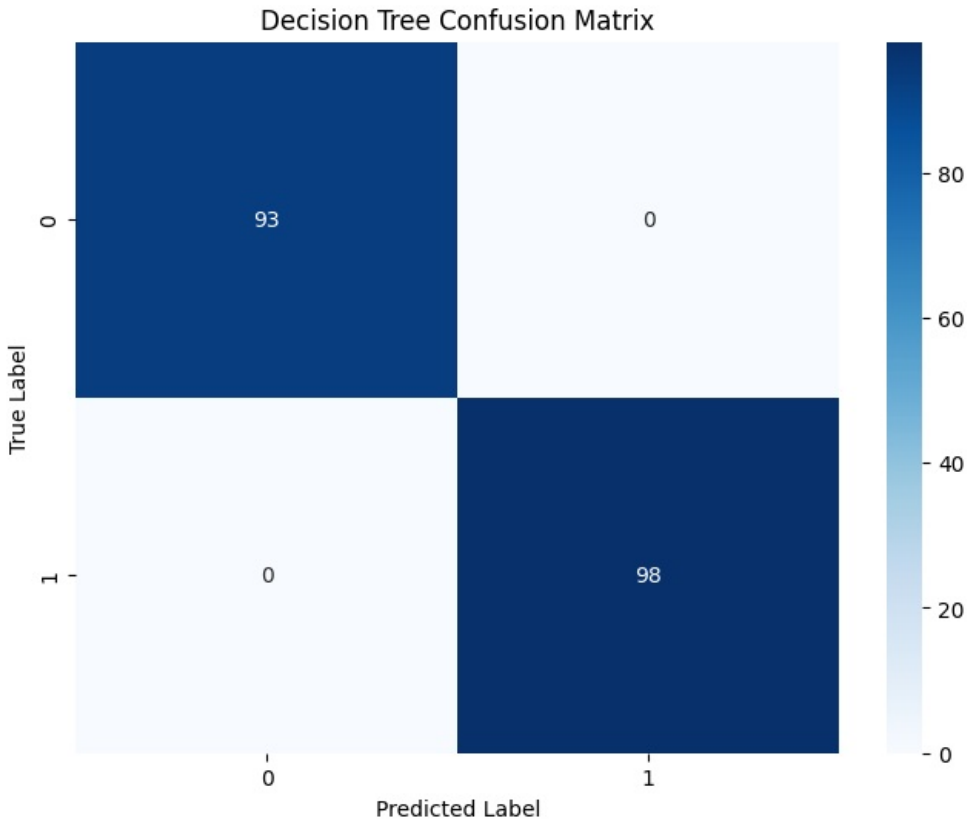
In [336…

```python
# Evaluate Decision Tree
dt_accuracy = accuracy_score(y_test_class, dt_pred)
print("\nDecision Tree Classifier Results (Predicting Purchased Bike):")
print(f"Accuracy: {dt_accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_test_class, dt_pred))
```

```
Decision Tree Classifier Results (Predicting Purchased Bike):
Accuracy: 1.0000

Classification Report:
              precision    recall  f1-score   support

          No       1.00      1.00      1.00        93
         Yes       1.00      1.00      1.00        98

    accuracy                           1.00       191
   macro avg       1.00      1.00      1.00       191
weighted avg       1.00      1.00      1.00       191
```

In [337…

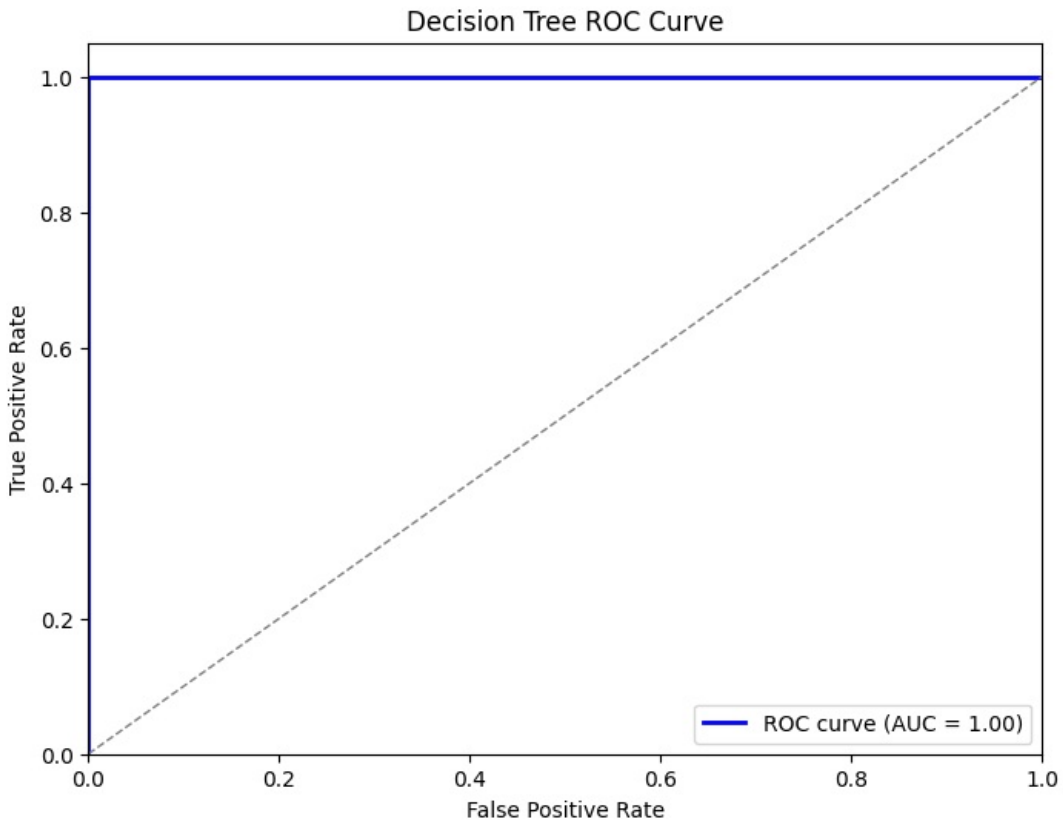
```python
# Confusion Matrix Visualization for Decision Tree
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test_class, dt_pred), annot=True, fmt='d', cmap='Blues')
plt.title('Decision Tree Confusion Matrix')
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.show()
```



In [338…

```python
# ROC Curve for Decision Tree
fpr_dt, tpr_dt, _ = roc_curve(y_test_class.map({'No': 0, 'Yes': 1}), dt_prob)
roc_auc_dt = auc(fpr_dt, tpr_dt)
plt.figure(figsize=(8, 6))
plt.plot(fpr_dt, tpr_dt, color='blue', lw=2, label=f'ROC curve (AUC = {roc_auc_dt:.2f})')
plt.plot([0, 1], [0, 1], color='gray', lw=1, linestyle='--')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Decision Tree ROC Curve')
```

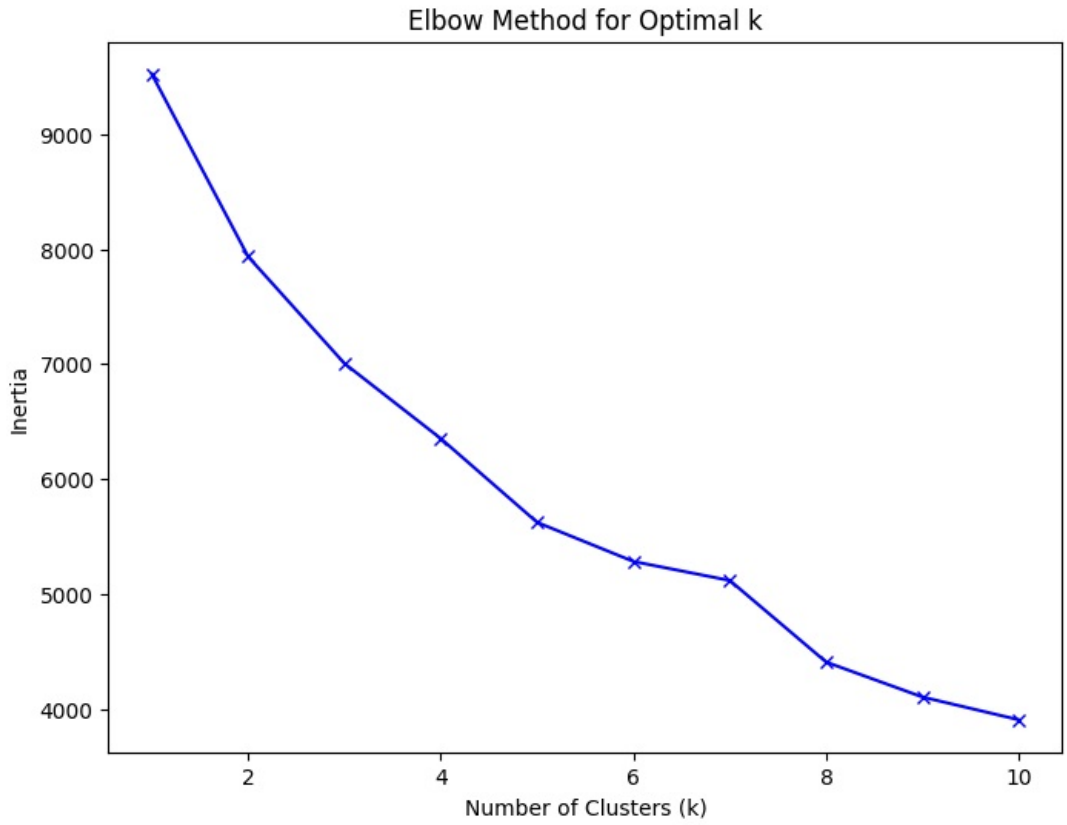
```
plt.legend(loc="lower right")
plt.show()
```



Clustering - For Hidden Patterns

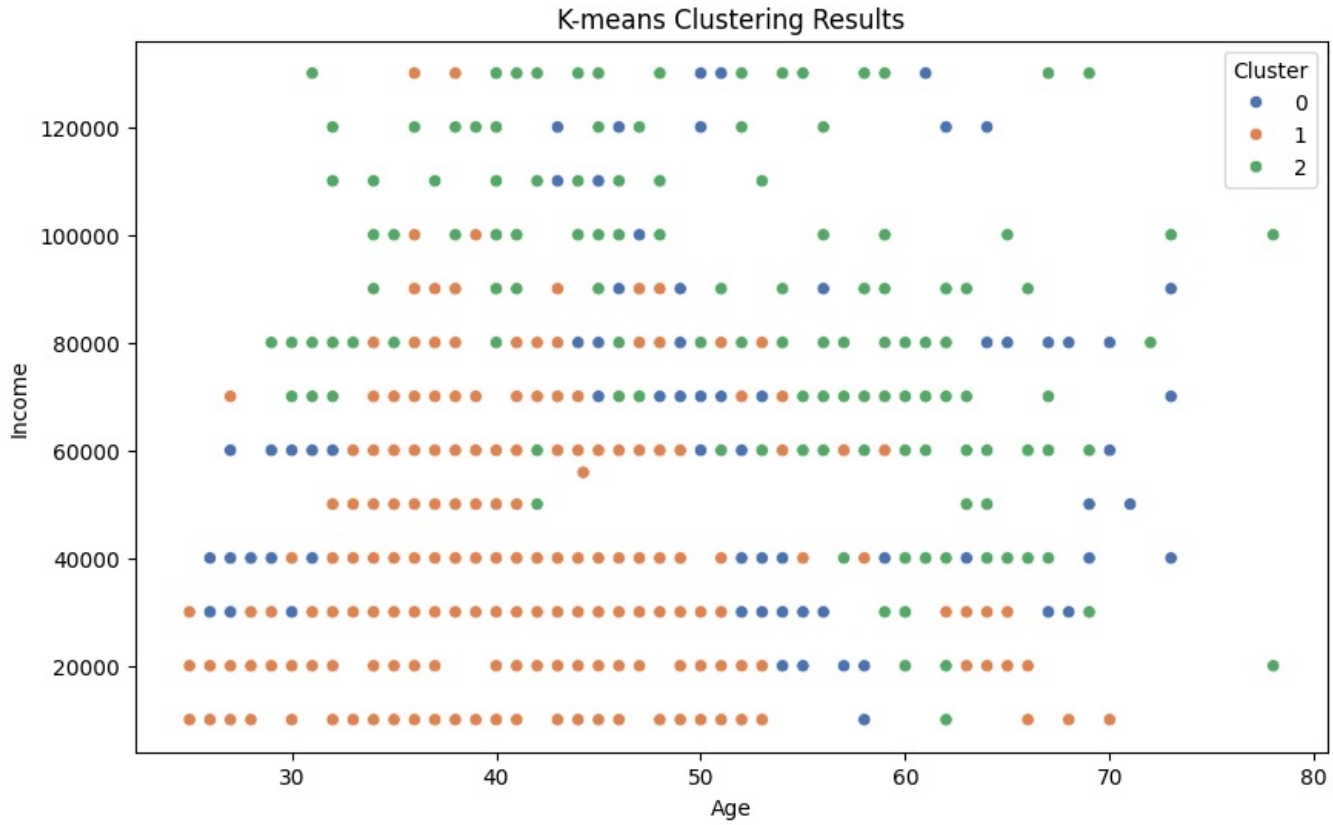
In [339…

```
# Scale the selected features for clustering
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Determine optimal number of clusters using elbow method
inertia = []
k_range = range(1, 11)
for k in k_range:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)

# Plot elbow curve
plt.figure(figsize=(8, 6))
plt.plot(k_range, inertia, 'bx-')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method for Optimal k')
plt.show()
```



In [340…

```
# Train K-means with optimal k (let's assume k=3 based on elbow curve)
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(X_scaled)
clusters = kmeans.predict(X_scaled)

# Add cluster labels to dataframe
dfd['Cluster'] = clusters

# Visualize clusters (using Age and Income as example, if available in selected features)
plt.figure(figsize=(10, 6))
if 'Age' in dfd.columns:  # Check if Age is still available
    sns.scatterplot(data=dfd, x='Age', y='Income', hue='Cluster', palette='deep')
    plt.title('K-means Clustering Results')
    plt.show()
else:
    print("Note: Age and Income not in selected features for visualization.")
```



In [341…

```
# Cluster analysis
```

In [341…

```python
# Cluster analysis
cluster_summary = dfd.groupby('Cluster').agg({
    'Income': 'mean',
    'Age': 'mean',
    'Children': 'mean',
    'Cars': 'mean'
}).round(2)
print("\nCluster Summary Statistics:")
print(cluster_summary)
```

```
Cluster Summary Statistics:
           Income    Age  Children  Cars
Cluster
0        58164.30  46.27      1.82  1.89
1        43120.03  40.16      1.43  0.84
2        79164.65  51.26      3.00  2.48
```

In [342…

```python
# Model Performance Interpretation
print("\nModel Performance Interpretation:")
print("1. Logistic Regression (Purchased Bike):")
print(f"- Accuracy of {log_accuracy:.4f} shows the proportion of correct predictions")
print("- See classification report for detailed precision, recall, and F1-score by class")

print("\n2. Decision Tree Classifier (Purchased Bike):")
print(f"- Accuracy of {dt_accuracy:.4f} shows the proportion of correct predictions")
print("- See classification report for detailed precision, recall, and F1-score by class")

print("\n3. K-means Clustering:")
print("- Cluster analysis shows distinct customer segments based on selected features")
print("- Visualize clusters to understand customer groupings (if applicable)")
```

```
Model Performance Interpretation:
1. Logistic Regression (Purchased Bike):
- Accuracy of 1.0000 shows the proportion of correct predictions
- See classification report for detailed precision, recall, and F1-score by class

2. Decision Tree Classifier (Purchased Bike):
- Accuracy of 1.0000 shows the proportion of correct predictions
- See classification report for detailed precision, recall, and F1-score by class

3. K-means Clustering:
- Cluster analysis shows distinct customer segments based on selected features
- Visualize clusters to understand customer groupings (if applicable)
```

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